

Automatic Enrolment and Household Saving-Instrument Portfolios: Evidence from Türkiye’s 2017 OKS Reform

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Abstract

Automatic enrolment reliably raises participation in retirement plans. Whether it raises household saving is a separate question, and plan-level data cannot answer it, because they observe one account rather than the household’s portfolio. We study Türkiye’s 2017 automatic-enrolment reform, which set eligibility by a worker’s age at an employer-specific inclusion date, using the Household Budget Survey. The survey records the instruments through which a household saves, up to four, from bank deposits and securities to gold, foreign exchange, real estate and investment in a household’s own business. We reconstruct legal eligibility from reported age and job tenure and compare eligible-type with ineligible-type households within the public sector over 2015–2019. Eligibility raises reported use of the pension-linked instrument by 33.9 percentage points. Non-pension participation falls by 0.5 points, but the confidence interval spans the ± 5 -point equivalence margin, so economically meaningful displacement cannot be ruled out. Overall saving participation rises by 12.2 points and portfolio breadth by 0.32 instruments, almost entirely through the pension-linked instrument itself. The reform thus changed which saving instruments households report. Consumption did not detectably fall, but the contributions involved, under 1% of household consumption on average, are too small for the survey to show how they were financed.

JEL codes: D14, G51, H55, J26

Keywords: automatic enrolment; pension saving; household portfolios; crowd-out; Türkiye

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1 Introduction

Automatic enrolment has become a common retirement policy tool, supported by extensive evidence showing that switching from an opt-in to an opt-out default raises participation quickly and substantially (Madrian and Shea, 2001; Choi et al., 2004). Whether the rise in participation results in proportional increases in household saving is a different and more challenging issue.

A household facing a new default has many ways to offset it. It can hold less in other instruments, contribute less elsewhere in the retirement system, run down liquid assets, borrow, consume less, withdraw balances, or change employer. Initially untreated workers may also catch up later by contributing at higher rates, thus narrowing the measured disparity without any actions being undone by the defaulted group (Choukhmane, 2025).

We examine whether a household reports using a specific saving instrument. The survey records which instruments a household possesses, but not the amount held in each. Amounts, balances, debt, withdrawals, and job histories are all unobserved, and every interpretation below is bounded accordingly. This boundary determines which of the above questions the evidence can answer.

Türkiye’s 2017 automatic-enrolment reform (*Otomatik Katılım Sistemi*, OKS) provides a helpful context for the question. Eligibility is based on statutory rules and age: employees younger than 45 at the relevant inclusion date were automatically enrolled, while those aged 45 and older were not. Eligibility is determined by a worker’s age relative to a dated regulatory event, not by the household’s saving choices. Public-sector implementation was staged by employer category, across two inclusion dates nine months apart. Enrolment cannot be directly observed in the survey data because the Household Budget Survey (HBS) does not include any variables related to OKS enrolment, opting out, or contributions in any wave. Nevertheless, reported age, length of employment, and social security registration allow us to bound each public-sector employee’s legal eligibility under each regulatory scenario. The reconstruction serves as our treatment variable, meaning each estimate reflects an intention-to-treat effect of legal eligibility rather than actual enrolment.

The value of the setting lies in the outcome side. HBS documents household saving instruments across the entire reported portfolio, not just a single provider or plan. In Türkiye, this portfolio is notably diverse: household saving runs substantially through gold, foreign exchange, real estate, and personal business investments, not standardised financial products (Temel Nalın, 2013). Pension-linked and non-pension instruments can be compared within the same households using identical specifications, addressing the participation versus saving question, which administrative plan records cannot accomplish. We explore whether being legally eligible for automatic enrolment affects household participation in saving instruments and the diversity of their portfolios, and what these changes reveal about substitution versus additional saving.

We conduct a repeated-cross-section difference-in-differences analysis within the public sector using data from HBS 2015–2019. Treatment is the reconstructed statutory eligibility described above, restricted to the invariant-classification sample, which includes households whose eligibility remains the same under both regulation-dated rollout scenarios. This ensures that group membership never relies on the one legal fact the survey cannot verify, representing 88.0% of public-sector analysis

households. The pre-period is 2015–2016 and the post-period 2018–2019. We exclude 2017 because the first inclusion date falls within that year and the survey does not record when households were interviewed. Standard errors are clustered by inferred birth cohort, with wild-cluster-bootstrap p-values reported as a companion inference check. The comparison involves eligible-type and ineligible-type households within the public sector. Therefore, the estimate reflects a relative effect compared to a comparison group that might also be indirectly exposed, and should not be interpreted as a lower bound.

We find, first, that legal eligibility raises reported use of the pension-linked instrument by 33.94 percentage points (95% CI [27.34, 40.55] pp). The outcome is a survey proxy rather than a direct measure of OKS enrolment. But the estimate is the most robust in the paper: precise under every inference method, stable across classification rules, and the only one whose non-zero conclusion survives the full trend-violation sensitivity grid. Second, participation in non-pension instruments changes by -0.50 percentage points, with a 95% interval of $[-10.25, 9.25]$ pp that spans both ends of the ± 5 percentage-point equivalence margin fixed before estimation, so extensive-margin displacement is neither confirmed nor ruled out. Third, overall participation in saving instruments rises by 12.22 percentage points (95% CI [2.13, 22.30] pp), a stable estimate with weaker causal attribution, and almost all of the increase comes from households whose only reported instrument is the pension-linked one. Fourth, portfolio breadth rises by 0.323 instruments on a 0–4 scale, almost entirely through the same instrument.

We also use income and expenditure data as a check on household resources. Neither disposable income nor consumption moves detectably, while the income–expenditure residual falls. A back-of-the-envelope calculation puts the contributions the default produces at about 1,000 lira a year per eligible household in 2016 prices if nobody opts out, and about 400 lira at the observed stay-in rate, under 1% of consumption. The residual’s fall is several times larger, too large to be explained by the payroll deduction alone, and the contributions are too small for the survey to show whether they were paid for by consuming less, from other saving, or by borrowing (Section 8).

The first finding is what the literature on defaults predicts. The effect of defaults on participation is among the most replicated results in household finance, and it extends to national automatic enrolment in the United Kingdom (Cribb and Emmerson, 2019, 2021) and to United States federal employees, the population closest to ours (Falk and Karamcheva, 2023). Chetty et al. (2014) explain why: most workers respond passively, so automatic contributions raise total saving with little offset, whereas price subsidies mostly move saving between accounts. Türkiye differs in one respect: opt-out was common. By the end of 2017, 54% of OKS contracts had been cancelled within the two-month window (Emeklilik Gözetim Merkezi (EGM), 2018), and at the end of 2018, 43% of public-sector contracts and 30% of private-sector contracts remained in force (Özyerden, 2026). A survey of private-sector employees links opting out to low trust in pension funds (Karayel, 2024).

Whether higher participation means more saving is less settled. Tax-favoured retirement accounts have been argued to create new saving (Venti and Wise, 1990; Poterba et al., 1995, 1996) and to reshuffle saving that households would have done anyway (Engelhardt, 2001; Attanasio and DeLeire,

2002). Benjamin (2003) finds both: about half of 401(k) balances are new private saving, and the reshuffling is concentrated among households that save the most. More recent designs find no detectable offset in other saving (Dolls et al., 2018; Goodman, 2020) but some adjustment through debt (Andersen, 2018; Beshears et al., 2022). The most complete evidence on net saving comes from register data: Jensen et al. (2025) observe Icelandic households’ full balance sheets and find that they did not reduce voluntary saving when mandatory contributions rose. This paper complements that evidence rather than competing with it. Comparable registers do not exist for Türkiye, and our survey cannot measure amounts. What it adds is the participation margin across a household’s whole portfolio, including gold, foreign exchange and real estate, in a setting where many enrolled workers opted out. Closest to our design is Lachowska and Myck (2018), who compare birth cohorts differently affected by a pension reform in the Polish household budget survey, although to estimate how public pension wealth affects saving, not participation.

Prior evidence on the Turkish reform comes from pension-company data. Yanıkkaya et al. (2023) find that net entry into the voluntary pension system fell after the reform, using monthly company-level flows without an untreated comparison group. Studies using the Household Budget Survey describe household saving portfolios but do not study the reform (Temel Nalın, 2013; Öksüz Nariç and Küçükönder, 2020; Dikmen, 2021). This paper provides household-level evidence on the reform: it reconstructs legal eligibility where enrolment is unobserved and follows the household’s whole saving portfolio instead of a single plan.

Section 2 describes the reform and Section 3 the data. Section 4 reconstructs eligibility and Section 5 sets out the design. Sections 6 to 8 report the results, the robustness checks, and the income and expenditure evidence, and Section 9 concludes.

2 Institutional Setting

2.1 The reform, eligibility, and rollout

Türkiye introduced automatic enrolment into the private pension system (*Otomatik Katılım Sistemi*, OKS) through Law No. 6740, adopted on 10 August 2016 and published in the Official Gazette (Resmî Gazete) on 25 August 2016, with effect from 1 January 2017. Under OKS, employers enrol their qualifying employees into a pension plan by default. Participation begins without an application, and continues unless the employee chooses to exercise the exit right described below. Two features of the legal design provide the identifying variation: a statutory age boundary that determines who is included, and a phased schedule by employer category that specifies timing.

Automatic enrolment applies to employees who have not yet reached age 45 at the relevant inclusion date. Employees aged 45 and above at that date are not automatically enrolled. The boundary depends only on the worker’s age at a dated regulatory event, not on any choice the worker makes or the worker’s saving behaviour.¹

¹A provision added to the regulation in December 2018 provides an alternative age-45 reference date of 1 January 2017 for inclusions predating that amendment. The invariant-classification sample of Section 4 gives the same treatment

The implementing regulation (annexed to Council of Ministers Decision 2017/9721, Official Gazette 2 January 2017, No. 29936) sets inclusion dates by employer category. Three provisions matter here:

1. Employees already working in public administrations listed in Schedules I–IV of Law No. 5018 were included from 1 April 2017.
2. Employees already working in other public administrations (public employers not listed in Schedules I–IV) were included from 1 January 2018.
3. Employees starting work with a public employer on or after that employer’s inclusion date are automatically included from their employment start date, provided they have not yet turned 45.

Private-sector employees were phased in separately by enterprise size over 2017–2019 and are not part of this paper’s analysis population.

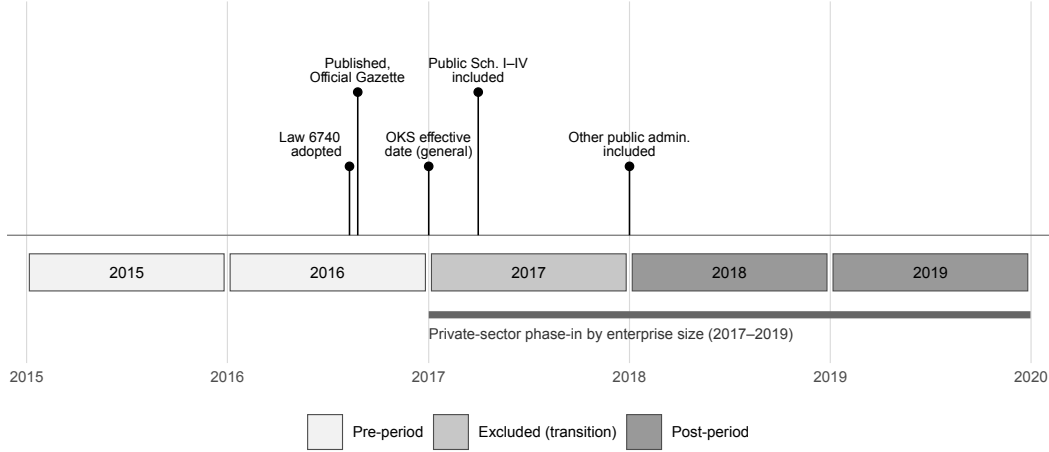
The public-sector timing matters for three reasons. First, the same legal rule takes effect nine months apart for two employer categories, so there is no single reform date for public-sector workers. Second, these categories are legal and budgetary classifications the survey cannot observe (Section 3.1), so the applicable date is unknown at the household level. The treatment reconstruction handles this by construction, not by assumption (Section 4). Third, administrative processing did not coincide exactly with the regulatory dates: bulk enrolment continued through 2018 and, according to industry monitoring accounts, ended only at the start of 2019. Legal eligibility, employer processing, and household awareness represent three separate timelines, with only the first being defined by regulation.

Figure 1 sets out these clocks together against the five Household Budget Survey waves used in the analysis. It shows the law’s adoption and publication in August 2016, the general effective date and the two staggered public-sector inclusion dates, and the private-sector phase-in band. Below them are the five equally represented survey waves from 2015 to 2019, with 2015–2016 as the pre-period, 2017 as the excluded transition year, and 2018–2019 as the post-period.

2.2 Opt-out and treatment interpretation

OKS provides a statutory opt-out right (*çayma*). Under the provision inserted by Law No. 6740, an employee "may withdraw from the contract within two months following the date on which they are notified of inclusion in the pension plan". Contributions and any investment return are refunded within ten business days. Exit remains possible after this period. Workers who opted out were subject to periodic re-enrolment, under rules that changed several times before the requirement was abolished at the end of 2021 (Sigortacılık ve Özel Emeklilik Düzenleme ve Denetleme Kurumu (SEDDK), 2022). A large share of the legally eligible population subsequently exits, so being enrolled

labels under either reading, because it requires agreement under the April 2017 scenario, whose reference year is also 2017.



Sources: Law No. 6740 (Resmî Gazete, 25 August 2016, No. 29812); implementing regulation annexed to Council of Ministers Decision 2017/9721 (Resmî Gazete, 2 January 2017, No. 29936), art. 5(2)-(3) and 6(2), for the public-sector inclusion dates and the private-sector phase-in.

Figure 1: Institutional timeline: reform dates and HBS survey waves

is not a permanent result of eligibility. This paper does not rely on any particular opt-out rate. Exit is part of the reform's design, however, and nothing below should be read as implying that eligible workers remained enrolled.

The same provision sets the contribution parameters. The employee contribution is 3% of earnings subject to the social security premium. The Council of Ministers (later the President) may double this rate, reduce it to 1%, or impose a fixed cap. The state contribution comprises a general rate, a one-off 1,000 Turkish lira for employees who do not opt out, and a further 5% of accumulated savings for participants who choose an annuity of at least ten years. None of these parameters enters any estimate. The contribution rate is given so that the reader can judge the size of the payroll deduction discussed in Section 8.

Three statuses need to be distinguished: legal eligibility for automatic enrolment is determined by age and employer category at a dated regulatory event; actual enrolment is determined by employer processing; and continued participation is determined by whether the employee opts out. The data record neither actual enrolment nor continued participation (Section 3.1). The treatment analysed here is legal eligibility alone. Every estimate below is an intention-to-treat (ITT) effect of *legal age-eligibility* and not a treatment-on-the-treated effect, a local average treatment effect, a participation-rate effect, or an effect conditional on enrolment. Because a substantial share of the eligible population may exit, the eligibility ITT is smaller than an effect conditional on continued participation would be, which these data cannot identify, and its size should not be read as a participation rate.

3 Data and Measurement

3.1 Data and sample

The analysis uses the Turkish Statistical Institute’s (TÜİK) Household Budget Survey (Hanehalkı Bütçe Anketi, HBS), survey years 2015–2019. HBS is an annual repeated cross-section: households are sampled independently in each wave and no household is followed over time. The 2022 wave is used only for the validation exercise below and is not pooled into the estimation sample. Fieldwork is continuous across the calendar year, with roughly one twelfth of the annual sample interviewed each month. The labelled survey year is the fieldwork year. HBS releases a household-level file and a member-level file, linked by a household identifier. Saving instruments, income, and expenditure are recorded at the household level, while age, employment status, job tenure, and the social security registration used as the sector proxy are recorded at the member level.

The analysis population is households in which at least one member is registered under the public-sector social security scheme (4C) and at least one member is a full-time salaried worker. No member of these households, working or not, is registered under the private-sector (4A) or self-employed (4B) schemes. Pooling 2015–2019 yields 4,057 household-years, containing 5,092 individual public worker person-year records, an average of roughly 1.26 identified public workers per household. Sector is identified from the member’s social security registration code (4A private, 4B self-employed, 4C public). Official documentation confirms that dependents are recorded under the registered worker’s code. A household’s sector composition can reflect one worker’s registration, while a formally employed private-sector member carries their own code.

The survey weight (**FAKTOR**) is defined at household level. Estimates are weighted, and unweighted estimates are reported alongside them with equal prominence. No primary sampling unit or stratum identifier is released in any wave used here or documented in any codebook, although the 2022 methodological publication describes the underlying design as stratified two-stage cluster sampling. Weighting therefore corrects the target quantity of the point estimate, but it does not give the standard errors design-based validity. This limitation cannot be repaired with the available data (Section 5.2).

The household saving module asks how the household holds the saving it has made, recording up to four instruments (**TASARRUF_1–TASARRUF_4**) chosen from a fixed list, with a distinct code for "does not save." Few households reach the limit: fewer than 1% report four instruments in any group and period, so the cap does not bind. The official reference period is the current status (*Cari*), and no recall window is specified in any codebook. The list is identical in every year 2015–2019: real estate, membership of a housing cooperative, gold, foreign exchange, bank account, equities, bonds/bills, fund participation certificate (option 8, discussed below), investment in one’s own business, lending money at interest, other, and does not save. The survey records which instruments a household uses, not how much it saves in each. No instrument-specific monetary amount exists in any wave. Every saving outcome in this paper is therefore an extensive-margin (participation) or count measure.

Four further gaps in the data shape the design. No variable records the interview month, date, or timing of fieldwork in either file in any year, so no household can be placed on a known side of a within-year regulatory date. No indicator of employer category distinguishes Schedule I–IV administrations from other public administrations, so the distinction determining the applicable inclusion date is unobservable at the household level. No birth date or birth year is recorded. Age (**YAS**) is whole years completed as of the survey month. Job tenure (**CALISMA_SURE_YIL**) is whole years, with tenure under six months coded zero. Neither has a month or day component. Finally, no administrative pension information is released. HBS records no OKS enrolment, opt-out, or contribution status, and no amount of premium paid to the voluntary private pension system (Bireysel Emeklilik Sistemi, BES) or to OKS. The questionnaire asks whether any member pays life insurance, individual pension, or voluntary insurance premiums, how many members do so, and the amount of individual pension premiums paid. Only the yes/no indicator is released, and the premium amount enters the microdata only within the aggregated block of non-consumption expenditure. The only pension variable with an amount is income received from a private pension (**GELIR_BIREYSEL_EMEKLILIK**). It records periodic receipts, not contributions, and because most private pension participants in Türkiye withdraw their accumulated savings as a lump sum, it is non-zero for only 161 individuals across the five waves, out of more than 200,000 member records. A full-text search of the official documents consulted (the household and member guides for 2015–2019, the questionnaire, and the 2022 methodological publication) returns no occurrence of "otomatik katılım" and no citation of Law No. 6740.

3.2 Saving outcomes and the option-8 measurement question

If automatic enrolment raises the probability that a household holds a pension-linked instrument, three things could be happening to the remainder of its portfolio. The household could add the pension instrument to an otherwise unchanged portfolio, raising overall saving participation. It could substitute the pension instrument for a non-pension instrument it previously used, leaving overall participation unmoved. Or both could occur in different parts of the population. The paper therefore examines four outcomes, reported in a fixed order. The order is interpretive, not a statistical gate: no result is suppressed or discounted because another outcome is or is not significant. The pension-linked participation proxy (**ips**) equals one if the household reports option 8 among its instruments. Non-pension participation (**saver_but_ips**) equals one if the household reports at least one saving instrument other than option 8, and serves as the extensive-margin displacement proxy. Any-saving participation (**saver**) equals one if the household reports any saving instrument at all, summarising net participation. Portfolio breadth (**savings_count**) counts the distinct instruments reported, 0 (not saving) to 4, and is secondary. None of these outcomes is monetary saving, a contribution amount, or wealth. They record whether an instrument is used, not how much is held in it, so monetary or intensive-margin crowd-out cannot be measured with these data. The term "displacement" below refers only to extensive-margin displacement (Section 6.3 states in full which margins are unobserved).

The pension-linked proxy is built on option 8. Reading option 8 as pension-linked rests on the timing of its rise, the 2022 change in the code list, and an official planning report, not on any TÜİK designation. This section presents the basis once, while subsequent sections include the resulting caveat without repeating the entire chain. Option 8's official label throughout 2015–2019 is "Fon katılım belgesi" (fund participation certificate), a generic investment-fund category. TÜİK documentation never uses "bireysel emeklilik" or "BES" for code 8 in any of these years, and no pension-specific option exists anywhere in the 2015–2019 code list. In 2022, TÜİK added two new codes, one for private pensions and one for digital currency, while code 8's label and position remained unchanged. There has been no official recoding of option 8.

Four further pieces of evidence concern the reading. First, the unweighted share of all households reporting code 8 in any of the four instrument slots is 0.23% (2015), 1.00% (2016), 1.65% (2017), 11.46% (2018), and 11.75% (2019). This jump at the reform boundary appears in no other code, including code 98 ("other"), which stays within a 5.5–6.6% band across all five years. Second, once an explicit pension code existed in 2022, code-8 selection fell to 0.34% (41 households) while 9.24% (1,101 households) selected the new pension code. Of those 41 code-8 households, 21 also selected the pension code, leaving a "code-8 only" share of 0.17%, at or slightly below the 2015 pre-reform baseline. Third, the committee report on consumption and saving trends prepared for the Twelfth Development Plan attributes the rise from 2018 in households holding saving in the fund participation category to the OKS regulations (Cumhurbaşkanlığı Strateji ve Bütçe Başkanlığı (SBB), 2023, p. 27). This is an official planning document's interpretation of the item, though not a TÜİK designation. Fourth, and pointing the other way, published Turkish work has used this item without a pension reading. Dikmen (2021) lists "fon katılım belgesi" verbatim among the 2019 responses and offers no pension interpretation, and Öksüz Nariç and Küçükönder (2020), modelling the same question on the 2017 wave, do not identify the response as a separate category.

Taken together, this is substantial, though not official, evidence for reading option 8 as a proxy for pension-linked saving activity in 2015–2019. The documentation shows that code 8 was never officially labelled as a pension category; the timing and the 2022 pattern point strongly to a pension reading; an official planning document reads the item the same way; and the published Turkish work warns against over-reading it. We therefore treat option 8 as a pension-linked saving proxy throughout, while stressing that the survey never designates it as a pension category. It is a measurement assumption with substantial supporting evidence, not a documented fact, and the caveat applies to every estimate that depends on option 8.

3.3 Income and expenditure

Annual household disposable income (YILKULGEL) has reference period "last one year" (a trailing twelve-month recall collected at the household's final interview visit). It is a constructed quantity: it sums cash and in-kind income components, adds imputed rent, applies a CPI index factor, and subtracts annual non-consumption expenditure and regular transfers to other households. Wage income enters net of tax and social-insurance deductions. Individual-pension premiums are an

enumerated item within the block of non-consumption expenditure that the construction subtracts.

Household expenditure (HARCAMA) covers the interview month, collected by diary during the fieldwork month, with durable purchases entered at one twelfth of value. It carries no index term. The two variables therefore differ in reference period, price treatment, and construction, which governs the interpretation of the mechanism module in Section 8.

4 Treatment Reconstruction

4.1 Worker-level classification

Treatment is legal eligibility for automatic enrolment. Establishing it for a sampled household requires four things the data do not directly provide. They are the worker’s age at a dated regulatory event; whether the worker was an incumbent at that employer on the inclusion date or a later entrant; which of two inclusion dates applies to that employer; and, for a multi-worker household, how member-level statuses aggregate. The reconstruction addresses each of these explicitly instead of collapsing them into an age cutoff. It is performed for each individual public-sector worker (full-time, currently working, registered under the public scheme), not for birth cohorts, since no birth date is observed.

The data provide only the survey year Y , whole-year attained age A , and whole-year job tenure T (needed only for 2016–2019, since 2015 precedes both inclusion dates). From these, three bounds are formed: age at a reference date in year R_y lies in $[(R_y - Y + A) - 1, (R_y - Y + A) + 1]$; the calendar year of the current job’s start lies in $[Y - T - 1, Y - T + 1]$; and age at that start lies in $[(A - T) - 1, (A - T) + 1]$. Each ± 1 -year band reflects a documented source of uncertainty: the unobserved interview month within the survey year, and the whole-year rounding convention in reported tenure. The bands are combined conservatively and are not assumed to offset each other.

A worker is classified definitely eligible if the entire applicable age bound lies below 45, definitely ineligible if its lower end is at or above 45, and ambiguous otherwise. Where the survey year precedes the inclusion year, incumbency is determined without tenure information. Otherwise, the job-start bound determines whether the incumbent or the later-entrant rule applies. A bound that straddles the inclusion year leaves incumbency ambiguous, but eligibility can still be definite if the two rules give the same answer. Full derivations and the decision procedure by case are in Appendix A.

4.2 Household aggregation and the invariant-classification sample

Because employer category is unobservable (Section 3.1), every worker is classified twice: once under the 1 April 2017 inclusion date (Schedule I–IV administrations) and once under the 1 January 2018 date (other public administrations). The two scenarios are not competing hypotheses to average over. Each worker belongs to exactly one real category, but the survey cannot say which. Estimating a full-sample model under either single date would assign every ambiguous household to one category by default. Pooling 2015–2019, the two scenarios classify the same household identically for 92.9% of household-years. Under the April 2017 scenario, the pooled shares are 61.8% definitely eligible,

29.8% definitely ineligible, 7.0% ambiguous, and 1.4% not classifiable. Under the January 2018 scenario, they are 58.2%, 33.3%, 7.0%, and 1.4%.

Household status aggregates member statuses in a fixed priority order, so that a household is labelled ineligible only if every worker is definitely ineligible, while a definitely eligible worker determines eligibility even if a co-worker is ambiguous (Section 7.1 tests the effect of that asymmetry). A household with at least one definitely eligible worker is eligible. Otherwise, one ambiguous worker makes it ambiguous. Failing that, one unclassifiable worker makes it unclassifiable. A household is ineligible only if every worker is definitely ineligible. This aggregation assumes that household saving can respond to the eligibility of any one member. Shared household budgeting makes this plausible, but the data cannot verify it, and it is treated as an identifying assumption (Section 5.3). Among multi-worker households in the estimation sample (27.3% of the sample), the two natural anchor choices, the youngest public worker and the member who determines eligibility, agree in every case, so the choice of anchor is not a source of arbitrariness.

The primary sample keeps only households classified the same way under both scenarios. A household is eligible if at least one public worker is definitely eligible under both, ineligible if every identified public worker is definitely ineligible under both, and excluded otherwise. This keeps 3,571 of 4,057 public-sector household-years (88.0%), pooled over 2015–2019. For these households, group membership does not depend on the one legal fact the survey cannot resolve. The excluded 12.0% (486 household-years) is kept as a labelled sensitivity population, not discarded. It is concentrated near the age-45 boundary and the rollout years and does not form a random subsample, which limits external validity to the full public-sector population (Section 9). Section 7.1 reports what can and cannot be estimated on it.

Using a single age cutoff for households (where the youngest member is under 45) would inaccurately represent the legal rule in four ways. It ignores the survey year, although the age bound at a fixed regulatory date depends on both survey year and attained age. It ignores the two inclusion dates, which shift that bound differently by survey year. It also ignores the later-entrant rule, under which eligibility is evaluated at the hire date. And it cannot represent multi-worker households, whose classification depends on the joint status of several individually bounded workers. That simpler rule agrees with the strict definite-status classification for 89–91% of household-years and disagrees outright for 0.7–2.1%. The remainder are cases the interval method leaves unresolved. Section 7.1 reports the estimate under it. Appendix A gives the case-by-case comparison.

Pre-period eligibility needs a further qualification. For waves preceding the applicable inclusion date, no legal eligibility fact yet exists for the sampled individual, and because HBS is a repeated cross-section, that individual is never observed again. Pre-period eligibility is therefore a "would-be-eligible type" classification: a fixed function of each wave's own characteristics, applied identically in every wave. It is not a claim that any specific household's members remained employed, stayed at the same employer, or were later enrolled. Post-period classification, by contrast, targets a real legal status that can be determined retrospectively. Its uncertainty comes from measurement precision, not from whether the fact exists yet.

5 Empirical Strategy

5.1 Specification

The primary specification is a within-public-sector, repeated cross-section difference-in-differences:

$$Y_{ht} = \alpha + \beta \text{Eligible}_h + \delta (\text{Eligible}_h \times \text{Post}_t) + \lambda_t + X'_{ht}\theta + \varepsilon_{ht}, \quad (1)$$

estimated by weighted least squares on household-years h in wave $t \in \{2015, 2016, 2018, 2019\}$. Eligible_h is the invariant-classification indicator of Section 4.2. $\text{Post}_t = 1$ in 2018–2019 and 0 in 2015–2016. λ_t is a full set of survey-year fixed effects, and X_{ht} collects household size, the household education measure, and a household sector-composition indicator (public, other public). The indicator is built from the member registration codes of Section 3.1. A household is coded "public" if every member with a registered scheme is registered under the public scheme, and "other public" if it also contains members not registered under any of the three employment-tied schemes (private, self-employed, or public). It is unrelated to the employer categories of Section 2.1, which the survey does not observe. δ is the coefficient of interest. The year fixed effects absorb the common movement in 2018 and 2019, so no separate Post main effect is needed. δ is estimated identically either way (Appendix A.7).

This specification (sample, treatment rule, timing, outcome set and order, weighting, covariates, functional form, inference method, and equivalence margin) was fixed before any outcome was estimated. The only outcome information consulted beforehand was pre-period (2015–2016) prevalence, used for the ex ante power calculation. The sensitivity exercise in Section 7 has a different chronology, stated there.

The pre-period is 2015–2016 and the post-period 2018–2019. 2017 is excluded from the primary contrast, not assigned to either side. Fieldwork spans the whole of 2017, the first inclusion date (1 April 2017) falls inside that year, and no observable variable places a household on either side of it, so coding 2017 as pre or post would impose a classification the data cannot support. The year is kept and reported separately as a transition diagnostic, and it enters the dynamic specification used for the sensitivity analysis in Section 7.

The covariate set is minimal, and each element is included for a stated reason. Household size corrects a documented composition imbalance (the ineligible group's mean household size falls by 0.27 over 2015–2019 while the eligible group's is roughly flat) and improves precision. Education is the completed-education level, on a five-point scale from no schooling to post-secondary, of one adult member selected by a fixed rule from the household panel (an eligible worker where one exists, preferring the household head; otherwise the oldest member). It is a precision and composition covariate and is not plausibly affected by treatment for an adult already in employment. The scale is unaffected by TÜİK's 2018 renumbering of the education codes, which maps one-to-one onto the same categories. Year and sector-composition indicators are standard. Household income is deliberately excluded. Automatic enrolment reduces enrolled workers' take-home pay through a payroll deduction, and the survey's disposable income variable is constructed net of such deductions.

Income is therefore a candidate mediator of the treatment and conditioning on it would block part of the channel being measured.

5.2 Weights and inference

Estimates are weighted. Unweighted estimates are reported with equal prominence, not as a secondary robustness check. Weighted and unweighted estimators coincide only under effect homogeneity across sampling-probability strata, which is not assumed. Effective-sample-size ratios, computed as $(\sum w)^2 / \sum w^2$ by group $\times$ year cell, lie between 0.70 and 0.79 in every cell of the primary sample, so no cell shows severe weight concentration.

Standard errors are clustered on birth cohort, constructed as survey year — attained age of each household’s eligibility-determining public worker (the definitely eligible worker where there is one, otherwise the youngest public worker). This is the level at which treatment is assigned. Households in the same cohort face the same age-specific labour-market and life-cycle conditions, and treatment status is constant within every cluster (56 clusters pooled, none with mixed treatment status; 55 clusters in the estimation sample). Since birth year is not observed, the cluster variable is a proxy built from whole-year age and survey year, with the same one-year uncertainty as the classification. Because the number of clusters is modest, headline inference is accompanied by a restricted (null-imposed) wild cluster bootstrap-t with Rademacher weights and $B = 9,999$ draws under a fixed seed (Cameron et al., 2008; Roodman et al., 2019), implemented in the `fwildclusterboot` package (Fischer and Roodman, 2021). Heteroskedasticity-robust (HC1) standard errors are reported only for comparability. They ignore the cluster dependence and are expected to be too small if within-cluster correlation is positive. Two degrees-of-freedom conventions appear in the reported quantities. Cluster-robust p-values use the model’s residual degrees of freedom, while the confidence intervals use the t distribution with $G - 1$ degrees of freedom. No significance conclusion changes under either convention. Cohort clustering addresses dependence at the level of assignment only. Spatial and survey-design dependence cannot be corrected without PSU and stratum identifiers, which are not released.

5.3 Identification

δ is an intention-to-treat effect of legal OKS age-eligibility on household saving-instrument outcomes, within the public sector, comparing eligible-type to ineligible-type households across the pre/post boundary. As Section 2.2 explains, it is not an effect of enrolment, contributions, or wealth. It is a relative effect against a comparison group that may itself be indirectly exposed. Ineligible-type public-sector households share workplaces with eligible colleagues and are exposed to the same publicity about the reform. Colleagues’ choices are known to affect retirement-plan decisions (Duflo and Saez, 2002, 2003). Direct contamination is not the issue. A scan of 1,210 comparison households found none containing a formally employed private-sector, self-employed, or uncounted public member, consistent with the construction of the sector variable. The 2.3% containing an economically active member without a formal-employment insurance code are in informal employment, which brings no

automatic-enrolment exposure. In 2018–2019, 18 comparison households contain a member under 45 who is coded to the civil-servant scheme but is not a full-time worker, mostly dependants in low-paid work. Their possible contribution is negligible. The remaining channel is indirect exposure through peers or publicity. It is part of what the estimand means, not a defect that sample construction can remove. Section 6.1 reports how the comparison group’s own outcomes moved over the period, which matters for reading δ .

δ should not be read as a lower bound in this paper. A lower-bound reading would require the spillover on the comparison group to be non-negative, not to reverse across outcomes or subpopulations, and not to be offset by bias from another source. None of these is established, and direct evidence warns against assuming a well-behaved comparison group. In a cross-sector specification using this same survey (Appendix D), public workers against the self-employed, both aged 45 and over, fail the parallel-trends test on the pension-linked outcome ($F = 3.15$, $p = 0.043$), the only failure among nine comparison-outcome cells tested.² Public versus self-employed is therefore not the primary comparison: self-employment is a structurally different labour-market position, so any sector-specific shock (Türkiye’s 2018 currency and macroeconomic disruption being the leading candidate) is observationally indistinguishable from a reform effect in that comparison. A triple-difference layering self-employed households on top of the within-public contrast would rest on a different untested assumption (no sector-specific age-differential trend), not a weaker one, with the one piece of direct evidence bearing on it pointing against it.

The main identifying assumption is parallel trends between eligible-type and ineligible-type public-sector households. Further assumptions are compositional stability of the repeated cross-sections, classification accuracy within the invariant sample, the household aggregation assumption of Section 4.2, and the absence of an age-threshold-specific differential trend unrelated to the reform. The last includes any age-specific differential response to the 2018 currency and macroeconomic disruption, which falls in the post-period, and year fixed effects absorb it only to the extent that it is common across ages. The parallel-trends evidence is limited. Excluding 2017 leaves the 2015 to 2016 change as the only observable pre-reform movement, and 2016 is legally pre-reform but post-announcement. The 2017 coefficient cannot fill the gap. It averages a partly realised treatment effect across households with different inclusion dates, so it is not a clean pre-treatment quantity, and a coefficient indistinguishable from zero shows only that no differential movement is detectable at this precision. No result in this paper is presented as validating parallel trends.

An age-45 regression discontinuity is not used, for reasons settled before any estimate was produced. The running variable would be discrete integer age, with documented mass points, and at the household level it is a constructed representative age that depends on a selection rule, and local support is thin. Most importantly, the invariant-classification sample excludes as ambiguous exactly the households whose classified worker sits nearest the legal boundary, which are the observations a sharp discontinuity design needs most. The narrow-age-band estimates of Section 7.3 provide the

²That test uses a different classification rule, pre-period, and covariate set from the primary design. Appendix D gives the comparison and reports it as evidence on the credibility of the cross-sector comparison itself.

age-local evidence instead, within the difference-in-differences framework.

6 Results

6.1 Sample and descriptive patterns

The estimation sample is built from 59,102 household-years in the full pooled 2015–2019 Household Budget Survey. Dropping the 55,045 household-years not exposed to the public sector leaves 4,057. Requiring agreement between the two rollout scenarios (Section 4.2) drops a further 486, leaving the invariant-classification sample of 3,571. Listwise deletion on the covariates and cluster variable drops 4 more, to a complete-case sample of 3,567. Setting aside the 708 household-years from 2017 leaves the estimation sample of 2,859: 963 pre-period eligible-type, 543 pre-period ineligible-type, 936 post-period eligible-type, and 417 post-period ineligible-type, spanning 55 birth cohorts. The four listwise-deleted rows are scattered across cells and do not materially affect any cell mean or count. Effective-sample-size ratios (n_{eff}/N , by group $\times$ year cell) range from 0.700 to 0.793, so no cell shows severe weight concentration.

Table 1: Sample composition, by treatment group and survey year

Year	Group	Households	Anchor age (mean)	Anchor age (median)	Household size (mean)	Secondary+ education	NACE: public admin.
2015	Eligible	403	33.9	34	3.31	97.5%	40.3%
2016	Eligible	561	34.0	34	3.33	97.2%	37.6%
2017	<i>Eligible</i>	<i>460</i>	<i>34.7</i>	<i>35</i>	<i>3.36</i>	<i>98.1%</i>	<i>40.5%</i>
2018	Eligible	453	34.8	34	3.22	99.0%	47.9%
2019	Eligible	484	35.5	36	3.28	99.1%	44.3%
2015	Ineligible	274	51.1	50	3.04	90.9%	38.9%
2016	Ineligible	270	51.7	51	2.98	90.9%	39.4%
2017	<i>Ineligible</i>	<i>249</i>	<i>52.9</i>	<i>52</i>	<i>3.03</i>	<i>91.0%</i>	<i>33.1%</i>
2018	Ineligible	224	54.0	53	2.92	94.0%	50.2%
2019	Ineligible	193	54.3	54	2.77	92.5%	34.1%
Pre (2015–16), pooled †	Eligible	964	34.0	—	3.32	97.3%	38.7%
Pre (2015–16), pooled †	Ineligible	544	51.4	—	3.01	90.9%	39.1%
Post (2018–19), pooled †	Eligible	937	35.2	—	3.25	99.1%	46.0%
Post (2018–19), pooled †	Ineligible	417	54.1	—	2.85	93.3%	42.7%

Diagnostics for the invariant-classification population ($N = 3,571$ household-years, before listwise deletion; 2017 shown for completeness though excluded from the estimation sample), calculated from pre-treatment and compositional variables only. No saving outcome enters this table. NACE: public admin. is the share of the group's full-time public-sector workers whose reported economic activity is public administration and defence (NACE Rev. 2 section O).

† Household-count-weighted average of the two year-level rows immediately above, computed from the cell means shown rather than re-estimated from microdata. Pooled counts (964/544/937/417) are computed before the four-row listwise deletion described above and so differ from the final estimation sample's 963/543/936/417 by at most one household per cell.

Table 2: Weighted group $\times$ year outcome means (percentage points; portfolio breadth in instruments)

Year	Group	Pension-linked	Non-pension	Any-saving	Portfolio breadth
2015	Eligible	1.10	38.73	39.83	0.456
2016	Eligible	3.98	44.53	46.03	0.539
2017	<i>Eligible</i>	<i>5.45</i>	<i>45.32</i>	<i>47.97</i>	<i>0.548</i>
2018	Eligible	53.65	52.80	75.49	1.138
2019	Eligible	51.99	55.15	75.72	1.179
2015	Ineligible	0.71	33.69	34.12	0.399
2016	Ineligible	3.14	33.60	34.36	0.424
2017	<i>Ineligible</i>	<i>4.43</i>	<i>41.65</i>	<i>43.61</i>	<i>0.497</i>
2018	Ineligible	20.45	43.35	53.04	0.750
2019	Ineligible	16.24	50.67	57.46	0.758

2017 is shown for transparency and is excluded from the primary contrast.

Table 1 shows the ineligible-type group’s drift in anchor age (+3.2 years over 2015–2019) and household size (−0.27). The eligible-type group is comparatively stable on both. The drift is gradual across the five years, not concentrated in one break, and we attribute it to no specific cause.

Table 2 reports the weighted group-by-year means of the four outcomes. These are unadjusted, survey-weighted sample means of the indicators, not model-predicted values. Pension-linked instrument use among eligible-type households rises from a 1–4% base in the pre-period to 52–54% in 2018–2019, roughly three times the rise among ineligible-type households.

Figure 2 plots Table 2’s three binary outcomes. In the pension-linked panel, both groups move together and barely at all through 2017, then diverge sharply in 2018. Here, the eligible-type line lands well above the ineligible-type line, but both rise from a base under 5%. The any-saving and non-pension panels show the more gradual divergence.

The comparison group’s own rise is large and requires discussion. Ineligible-type households move from 0.71% (2015) and 3.14% (2016) to 20.45% (2018) and 16.24% (2019), a rise of roughly 13 to 17 percentage points off a very low base, in a group the statute excludes. It is not direct contamination, since no household in this group contains a formally employed private-sector member, a self-employed member, or a public worker classified as eligible under either rollout scenario. It also does not show that the classification leaks: workers under 45 at inclusion but older at interview are placed in the ± 1 -year straddle band and are excluded as ambiguous. The national all-household code-8 share (11.46% in 2018) is not a useful benchmark either: it pools a population with far lower formal employment and financial participation, and the eligible group’s own 52–54% exceeds it by much more.

Four explanations are consistent with the pattern, and we cannot choose among them. Spillover and salience could induce voluntary individual-pension take-up or changed reporting among the ineligible, through workplace proximity to enrolled colleagues and the reform’s general visibility. Age and cohort composition may also matter: voluntary pension saving rises with age above 45 (Guariglia and Markose, 2000), and the comparison group’s mean worker age rises 3.2 years across the window while the eligible group’s is comparatively stable. A repeated cross-section does not hold group composition fixed. Measurement is a third candidate, since code 8 is officially a generic fund participation certificate. Any rise in genuine investment-fund holding, or any reporting of

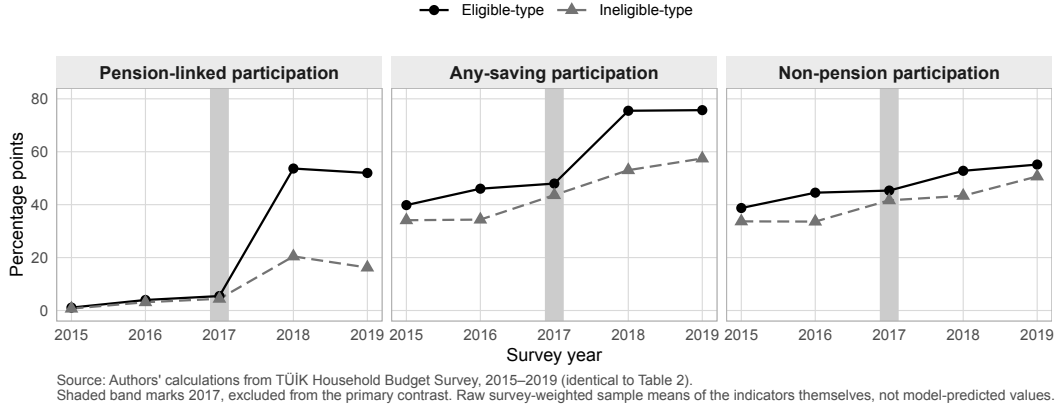


Figure 2: Weighted group $\times$ year outcome means, 2015–2019

a member's arrangement, a lapsed plan, or a plan held by someone outside the classified worker set, enters this variable in the same way. And a broader contemporaneous change is a fourth: the national code-8 share rose more than sixfold between 2017 and 2018 across all households, a movement that includes the private-sector phase-in and is not confined to this design's population. On the first explanation, Yanıkkaya et al. (2023) report a negative significant automatic-enrolment coefficient in their 45-and-over column. Net new entry into the voluntary individual pension system fell after the reform among participants above the eligibility boundary, which the authors describe as an indirect effect on the ineligible. Like our finding, this shows that the statutorily ineligible population's pension behaviour did not stay flat. But it is a national monthly flow of voluntary entrants at the pension-company level, from a design its own authors call mostly descriptive, while ours is a household current-status report. Two qualifications limit what this comparison can show. Their dependent variable is net entry, and terminations in the voluntary system also rose over the period (EGM data reported in Özyerden, 2026, p. 90), so gross entry among the over-45s could have risen while net entry fell.

Separately, the mean age of new voluntary entrants rose by about a year between 2016 and 2019, from 36.6 to 37.7 for men and from 37.4 to 38.5 for women (EGM data reported in Özyerden, 2026, p. 91). That is consistent with older workers entering voluntarily, but also with younger would-be entrants being diverted into OKS, and it does not distinguish public-sector from other workers. Neither piece of evidence supports or rules out a voluntary take-up explanation for our comparison group's rise. Both are context, not a test.

This pattern determines how each coefficient below should be read. The difference-in-differences estimate is a relative effect of legal eligibility, measured against a comparison group whose own reported use of the pension-linked instrument rose substantially over the same period. It is not an estimate of the reform's total effect relative to a wholly unexposed population, and no such estimate is available from these data. Nor is it a lower bound. Under the measurement and contemporaneous-change explanations, part of the eligible group's rise is the same non-behavioural

movement, which the difference-in-differences nets out by design, so the relative estimate could lie above or below any total effect.

6.2 Primary estimates

Weighting changes no magnitude or inferential conclusion for any of the four outcomes. The non-pension estimate changes sign between the weighted and unweighted versions, but both values are small and statistically indistinguishable from zero. Section 8 reports a mechanism outcome for which weighting does change a conclusion. The wild-bootstrap p-value for the pension-linked outcome is reported as below the simulation floor $1/(B+1) \approx 0.0001$, not as zero.

Table 3: Primary difference-in-differences estimates, weighted and unweighted

Outcome	Weighted	Cluster SE	95% CI (cluster)	Wild-boot. p	Unweighted	Cluster SE (unw.)
Pension-linked	+33.94 pp	3.30 pp	[27.34, 40.55] pp	<0.001	+33.94 pp	2.76 pp
Non-pension	−0.50 pp	4.86 pp	[−10.25, 9.25] pp	0.916	+0.22 pp	4.38 pp
Any-saving	+12.22 pp	5.03 pp	[2.13, 22.30] pp	0.026	+13.30 pp	4.44 pp
Portfolio breadth	+0.323 instr.	0.081	[0.160, 0.486] instr.	<0.001	+0.338 instr.	0.070
<i>implied pension-only</i> †	+12.72 pp	—	—	—	+13.07 pp	—
<i>implied non-pension count</i> †	−0.016 instr.	—	—	—	−0.001 instr.	—

N = 2,859, 55 birth-cohort clusters. Coefficient on Eligible $\times$ Post. The last two rows are accounting decompositions computed by exact subtraction of the rows above, not separately estimated coefficients; they carry no standard error, confidence interval, or p-value.

† Accounting decomposition, defined below: pension-only = any-saving − non-pension; non-pension instrument count = portfolio breadth − pension-linked. Not pre-specified, and no inference is attached. Cluster-robust p-values, alongside the wild-bootstrap p-values shown (conclusions match at the 5% level throughout): pension-linked <0.001; non-pension 0.918; any-saving 0.015; portfolio breadth <0.001.

The four outcomes are not independent measurements. Define the pension-only indicator as one when a household reports option 8 and no other instrument. Non-pension participation and the pension-only indicator are mutually exclusive and together cover all savers, so any-saving participation equals their sum. Likewise, portfolio breadth equals the pension-linked indicator plus the count of other instruments. Because all four outcomes are estimated on the same sample with the same right-hand side and weights, the Eligible $\times$ Post coefficients add up exactly, and the components follow by subtraction. The pension-only component is +12.72 pp weighted (+13.07 unweighted). The non-pension instrument count is -0.016 instruments weighted (-0.001 unweighted). These are accounting decompositions of the point estimates. They carry no standard error or p-value, because the variance of a difference requires the covariance between the two estimators, which is not recovered here, and no inferential claim is made about either component.

Two facts follow, and Section 6.3 relies on both. First, the estimated increase in overall saving-instrument participation is concentrated almost entirely in households whose only reported instrument is the pension-linked one. Second, the estimated increase in portfolio breadth is, to within 0.016 instruments, the pension instrument itself. Non-pension breadth is essentially unchanged.

6.3 Interpreting the estimates

For pension-linked participation, legal eligibility raises the probability that a household reports the pension-linked instrument by 33.9 percentage points (95% CI [27.3, 40.6] pp). This is a relative intention-to-treat effect of legal eligibility on reported use of the pension-linked saving proxy. The estimate is large and precise, identical to two decimal places under weighted and unweighted estimation, and significant under HC1, cohort-clustered, and wild-bootstrap inference. It is not an artefact of covariate adjustment, since reconstructing the difference-in-differences directly from the raw weighted group means in Table 2 gives approximately $(0.528 - 0.028) - (0.183 - 0.019) \approx 0.337$, against the adjusted 0.339.

The measurement caveat of Section 3.2 applies: the estimate indicates that eligible-type households became roughly 34 percentage points more likely to report using the pension-linked saving instrument, relative to ineligible-type households, under a triangulated but not officially confirmed reading of option 8. It does not say that 34% of eligible households enrolled in OKS, and implies nothing about enrolment, contributions, balances, wealth, or net saving. Statistical robustness and measurement validity are also different questions. That the coefficient is large, precise, and stable across specifications shows that the estimate is not an artefact of design choices. It does not show that option 8 measures what we read it as measuring. The post-period level is also high: 53.65% of eligible-type households report the instrument in 2018 and 51.99% in 2019. The survey does not record enrolment. But EGM data on public-sector contracts still in force, 43% at the end of 2018 and 37% at the end of 2019 (Özyerden, 2026), predict that 37–49% of eligible households held at least one contract, plus the 3% who reported the instrument before the reform.³ The range depends

³With p the share of contracts in force (43.2% for 2018, 37.1% for 2019) and k the number of eligible workers in a household (one in 79% of eligible households in 2018 and 74% in 2019, two in nearly all others), the share of

on whether workers in the same household decide jointly or independently. The observed level is close to the top of that range in 2018 and about 6 points above it in 2019, consistent with the option-8 reading. But the comparison group’s own rise shows that option 8 also captures holdings other than OKS contracts, such as voluntary private pensions or investment funds.

For non-pension participation, the point estimate is -0.50 percentage points ($p = 0.918$; 95% CI $[-10.25, 9.25]$ pp). The unweighted estimate is $+0.22$ percentage points, with a confidence interval of $[-8.56, 9.01]$ pp. The equivalence margin for this outcome was fixed before estimation at ± 5 percentage points. This margin was chosen on substantive grounds, as roughly an eighth of the outcome’s own pre-period base of about 42% and a movement large enough to matter for the substitution question. It was deliberately not derived from the design’s ex ante minimum detectable effect (10.7–12.3 pp). Equivalence is declared only if the entire 95% cluster-robust confidence interval lies inside the margin, which is equivalent to a two one-sided-tests procedure at $\alpha = 0.025$ and stricter than the conventional $\alpha = 0.05$ (Schuirmann, 1987). Here the interval spans both margins in the weighted and unweighted specifications, so the equivalence criterion fails: this design cannot rule out extensive-margin non-pension displacement of policy-relevant size in either direction. The point estimate is close to zero, but failing to establish equivalence shows neither that displacement is absent nor that it is present. The result is inconclusive, and the interval, not the point estimate, is the summary of what the data say. Gelber (2011) reports 401(k) eligibility results in the same way: no significant effect on other financial assets, with intervals too wide to rule out substantial changes.

Because "crowd-out" can mean several distinct economic objects, we state which one this coefficient bounds. It concerns whether an eligible household becomes less likely to report at least one non-option-8 instrument, an indicator that moves only when a household crosses between holding some non-pension instrument and holding none. It says nothing about amounts: reductions in holdings where participation continues, lower voluntary contributions elsewhere in the pension system, withdrawals, debt-financed contributions, changes in consumption, substitution between assets and liabilities, or adjustments through job changes. The survey records no amount for any instrument and no household liabilities, so these are not gaps a different specification would close. The contrast with Jensen et al. (2025) makes the boundary concrete. Their crowd-out estimand is monetary and derived from a full balance sheet, so their design can see the amount margins that ours cannot. Their result is therefore a prior about how households behave in a comparable setting, not evidence about our interval, and our near-zero point estimate is not a smaller-scale version of their finding, since the two coefficients measure different things. The result, stated exactly, is that no extensive-margin displacement is statistically detectable, economically meaningful displacement cannot be ruled out, and the monetary and balance-sheet margins are unobserved. The paper does not claim that there was no crowd-out or substitution, or that additional saving has been established.

For any-saving participation, eligible households’ overall saving-instrument participation rises by

households with at least one contract is p if workers decide together and $1 - (1 - p)^k$ if they decide independently: 43–49% in 2018 and 37–43% in 2019. Adding the 2.8% pre-reform base assumes those households are not the same ones. The benchmark is illustrative.

12.22 percentage points (95% CI [2.13, 22.30] pp), significant under every inference method used (HC1 $p = 0.004$; cluster $p = 0.015$; wild-bootstrap $p = 0.026$). The unweighted companion is +13.30 pp. The point estimate is stable. What is weaker is the evidence that it reflects a causal effect rather than a pre-existing differential trend. By the identity of Section 6.2, the 12.22-point rise decomposes into +12.72 pp in households whose only reported instrument is option 8 and -0.50 pp in non-pension participation. As an accounting matter, the increase in any-saving participation is matched by the rise in pension-only reporting relative to the comparison group; the survey does not follow households, so individual transitions are not observed.

Any-saving participation does not depend on how option 8 is classified: it records whether a household reports any saving instrument, and its construction would be the same whatever code 8 represents. However, since the increase is concentrated in households whose only instrument is option 8, the economic interpretation of the 12.2-point rise depends on what option 8 represents, and the estimate does not independently establish additional financial saving. If option 8 is read as pension-linked, the rise describes households brought into reporting a saving instrument by the default: a participation change, not a monetary one.

The single available pre-period movement is not small relative to the estimate. The raw gap between eligible and ineligible groups in any-saving participation widens by 5.97 percentage points between 2015 and 2016 (weighted; 4.73 pp unweighted), just under half the post-period coefficient. The covariate-adjusted analogue is +7.62 pp with a standard error of 5.72 pp ($p = 0.183$). A wide, non-significant pre-trend coefficient shows that this single transition is underpowered to detect a differential trend, not that none exists, and that transition is itself partly post-announcement. The decomposition identifies this concern: of the +5.97 pp change in the raw any-saving gap between 2015 and 2016 (year-specific coefficient +7.62 pp), +5.89 pp (+7.26 pp) is in non-pension participation and only +0.07 pp (+0.35 pp) in the pension-only component. Pension-linked participation itself moves +0.46 pp (+0.81 pp). The pre-period movement is therefore almost entirely in the non-pension component, while the post-period effect is almost entirely in the pension-only component. This does not eliminate the identification concern, since a differential trend in non-pension participation would still bias the any-saving estimate, but the concern is not located where the estimated effect is. Two more qualifications are also relevant. This result is the most sensitive of the four outcomes to the inference method, with the p -value moving from 0.004 under HC1 to 0.026 under the wild bootstrap though the conclusion at the 5% level is unchanged. The quantity is participation rather than total household saving. A household that starts holding a new instrument moves this indicator regardless of how much it saves. Conversely, a household that saves more through instruments it already holds does not affect this indicator.

Finally, portfolio breadth, the count of distinct saving instruments used, rises by 0.323 instruments (95% CI [0.160, 0.486] instruments) on a 0–4 scale. It is a count, not a proportion, and it is never presented on a shared axis with the three participation measures. By the second identity of Section 6.2, the increase is the pension instrument itself: the implied change in the count of non-option-8 instruments is -0.016 instruments (weighted; -0.001 unweighted). Taken with the 12-point

Table 4: Year-specific Eligible $\times$ year coefficients (weighted, HC1, 2015 base)

Outcome	2016	2017	2018	2019
Pension-linked	0.008 (p = 0.68)	0.008 (p = 0.70)	0.331 (p < 0.001)	0.358 (p < 0.001)
Non-pension	0.073 (p = 0.20)	-0.008 (p = 0.90)	0.053 (p = 0.40)	0.013 (p = 0.84)
Any-saving	0.076 (p = 0.18)	-0.006 (p = 0.92)	0.177 (p = 0.004)	0.147 (p = 0.017)
Portfolio breadth	0.077 (p = 0.33)	0.0003 (p = 0.997)	0.344 (p < 0.001)	0.388 (p < 0.001)

increase in any-saving participation and an almost zero displacement estimate, this is consistent with households adding an instrument to an otherwise unchanged portfolio rather than substituting. Because the increase is essentially the pension instrument alone, portfolio breadth is best read as pension-linked participation restated on a count scale, not as independent corroboration, and it inherits the option-8 caveat.

6.4 Year-specific pattern

Table 4 presents the specification with Eligible interacted with year indicators (2015 base, all five waves, HC1 standard errors). Every 2017 coefficient is small and indistinguishable from zero, consistent with the inclusion dates falling partway through or after that fieldwork year. This shows that no differential movement is detectable in the transition year, not that no anticipation occurred.

The 2018 and 2019 coefficients appear to move in different directions across outcomes: pension-linked participation and portfolio breadth are larger in 2019 than in 2018, while any-saving and non-pension participation are smaller. The decomposition of Section 6.2 shows this is one movement. Every pension-linked quantity rises from 2018 to 2019 (pension-linked participation by 2.69 pp, the pension-only component from 12.47 pp to 13.40 pp, portfolio breadth by 0.043 instruments), while non-pension participation falls by 3.96 pp. Any-saving participation, the sum of the two components, therefore declines on net. This is arithmetic, not a behavioural explanation, and it does not alter the pooled 2018–2019 estimates in Table 3, which are the paper’s primary results.

7 Robustness and Identification Sensitivity

The classification and functional-form checks of Sections 7.1 and 7.2 were specified in the pre-estimation design record, in the form run here, before any primary coefficient existed. The narrow-band and trend-violation exercises of Sections 7.3 and 7.4 differ in one respect that affects how their evidence should be weighted. The methods were named there as optional robustness checks, but their band widths and the sensitivity grid were fixed only after the primary and first robustness results were known, though before the sensitivity estimates were computed. The parameters were chosen on design grounds (age comparability against cluster count and effective support), and the candidate band widths had been tabulated beforehand from age and weight fields alone, without reading any outcome. They are not pre-specified in the sense that the primary model is, and this section does not describe them as such.

Table 5: Eligible $\times$ Post under alternative treatment classifications and functional form

Specification	Pension-linked	Non-pension	Any-saving	Portfolio breadth
Primary (weighted, N = 2,859)	0.339	−0.005	0.122	0.323
Symmetric-ambiguity sample (N = 2,782)	0.341	−0.002	0.125	0.329
Single-date classification rule (N = 3,241)	0.292	0.023	0.126	0.303
Logit average cross-difference (unweighted, N = 2,859)	0.306	−0.003	0.125	—

7.1 Classification robustness

The symmetric-ambiguity sample also excludes any households with an ambiguously classified worker, eliminating the slight asymmetry where the priority rule allows an ambiguous co-worker in eligible households but not in ineligible ones. Every coefficient moves by less than 0.006, with no sign or significance change.

The single-date rule replaces the interval classification with the simpler age-based rule of Section 4.2, holding timing, covariates, weights, and clustering fixed. Pension-linked participation shows the largest movement of any check, falling by 4.8 pp to 0.292. It remains significant by a wide margin, and its confidence interval ([0.224, 0.359]) substantially overlaps the primary interval. Any-saving participation and portfolio breadth move slightly (to 0.126 and 0.303, respectively). Non-pension participation changes sign, but both values are small, and neither is significant.

Re-estimating on the excluded rollout-sensitive subgroup under each regulatory scenario is not informative. Under the April 2017 date, 146 of the 486 excluded household-years are definitively classified, all as eligible. Under the January 2018 date, 142 are definitively classified, all as ineligible. Neither subgroup contains treatment variation within its scenario, so the difference-in-differences contrast cannot be estimated there.

7.2 Functional form

As a functional-form check, the three binary outcomes are re-estimated by logit and reported as the average discrete cross-difference in predicted probabilities, the sample-averaged interaction effect in the sense of Ai and Norton (2003), instead of the logit interaction coefficient. That coefficient is not the interaction effect in a nonlinear model.⁴ All three preserve the sign and the qualitative significance conclusion of the linear specification (Table 5).

This column asks only whether the qualitative pattern survives a bounded link function. It is not identification robustness, since a logit specification rests on a parallel-trends assumption on the log-odds scale, which differs from and is generally incompatible with the level-scale assumption behind the linear headline. The logit is run unweighted and compared with the unweighted linear estimate, because the logit did not converge with the survey weights used as prior weights. No alternative estimator was substituted. Appendix B gives the details.

⁴Puhani (2012) shows that in a nonlinear difference-in-differences model the treatment effect is the incremental effect of the interaction coefficient, not the cross difference. The column reported here is the cross difference, used only as a functional-form check, and it does not estimate the treatment effect.

Table 6: Narrow-age-band estimates (weighted)

Outcome	Full sample (N = 2,859, 55 clusters)	± 10 years (N = 1,555, 22 clusters)	± 5 years (N = 707, 12 clusters)
Pension-linked	+33.94 pp [27.34, 40.55]	+27.59 pp [20.55, 34.63]	+23.71 pp [18.04, 29.38]
Any-saving	+12.22 pp [2.13, 22.30]	+16.53 pp [3.31, 29.75]	+12.51 pp [-5.23, 30.24]
Non-pension	-0.50 pp [-10.25, 9.25]	+6.31 pp [-7.44, 20.05]	+0.71 pp [-11.67, 13.10]

7.3 Narrow-age-band sensitivity

Restricting the sample to households whose anchor worker’s age lies within a fixed band around 45 tests whether the estimates are driven by age composition, by comparing eligible and ineligible households close in age. Two widths were fixed in advance of estimating any narrow-band coefficient: ± 10 years (primary) and ± 5 years (secondary) (Table 6).

Every narrow-band interval overlaps the corresponding full-sample interval. Pension-linked participation remains large and highly significant in both widths, declining only modestly to 81% and 70% of the full-sample magnitude as the comparison narrows. The any-saving point estimate does not collapse toward zero: at ± 10 it is somewhat larger than in the full sample, which weighs against an age-composition artefact. But precision falls sharply, and the ± 5 estimate is not significant (cluster $p = 0.121$). Non-pension participation remains near zero and non-significant at both widths, with intervals that continue to span or nearly span the ± 5 pp equivalence margin.

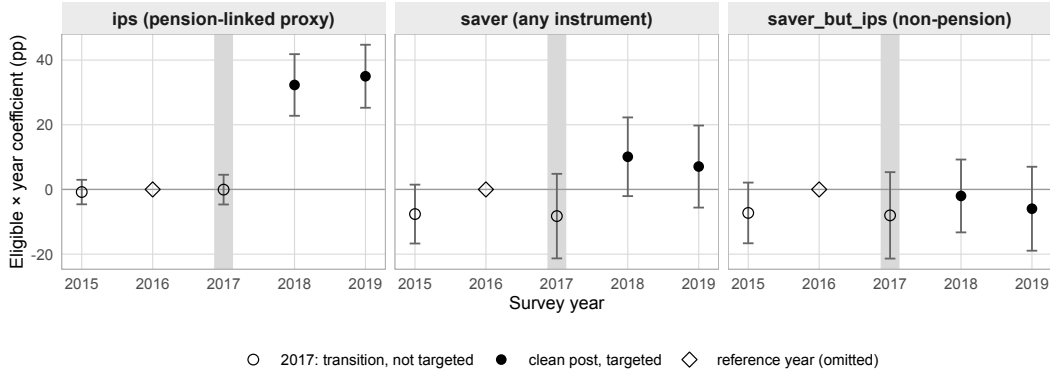
Inference at the ± 5 width is coarser. With 12 clusters, the wild bootstrap fully enumerates all $2^{12} = 4,096$ Rademacher sign vectors, and because a sign vector and its negative give the same absolute test statistic, only $2^{11} = 2,048$ distinct two-sided p -values are attainable. No alternative weighting or clustering was substituted in response.

7.4 Dynamic companion and trend-violation sensitivity

Sensitivity to violations of parallel trends is assessed using the relative-magnitudes restriction of Rambachan and Roth (2023), implemented with the `HonestDiD` package (Rambachan and Roth, 2026). That restriction bounds each period-to-period change in the differential trend after treatment by a factor $\bar{M}$ of the largest such change observed before treatment. It is applied to a dynamic companion specification: Eligible interacted with year indicators, all five waves, 2016 as the omitted reference, cohort-clustered covariance, fitted for this purpose alone. The grid $\bar{M} \in \{0.5, 1, 1.5, 2\}$ was fixed in advance of computing any bound. The targets are 2018 and 2019.

The exact form of the restriction here matters. The event-study vector contains one pre-period coefficient (2015, relative to the 2016 reference) and three post-period coefficients (2017, 2018, 2019). The single observed pre-treatment change is therefore $|\delta_{2015}|$, and the restriction bounds each subsequent step by $\bar{M} \cdot |\delta_{2015}|$. Because 2018 is two steps from the reference and 2019 is three, the permitted violation in levels accumulates: up to $2\bar{M} \cdot |\delta_{2015}|$ at the 2018 target and $3\bar{M} \cdot |\delta_{2015}|$ at the 2019 target. The exercise is thus more permissive than a one-step reading would suggest, and the confidence sets also account for δ_{2015} being itself estimated, not known.

Figure 3 plots the dynamic companion’s coefficients (tabulated in Appendix B, Table B4). For



Source: Authors' calculations from Tlk Household Budget Survey, 2015–2019. Dynamic companion specification of Section 7.4 (N = 3,567, 56 clusters), not the pooled primary estimand of Table 3. 2016 is the omitted reference (fixed at 0). 2017 is shaded and shown as an open circle: a transition-year coefficient, not a clean pre- or post-treatment quantity.

Figure 3: Dynamic companion coefficients, 2016 reference, cohort-clustered 95% CI

ips, a small, tightly estimated 2015 coefficient and a near-zero 2017 coefficient are followed by large, precisely estimated 2018 and 2019 coefficients (+32.29 pp and +34.99 pp) whose intervals do not approach zero. For **saver** and **saver_but_ips**, every interval crosses zero.

For **ips**, zero lies outside the robust confidence set at every grid value including $\bar{M} = 2$, so the sample breakdown value is strictly greater than 2. Concretely, with $|\delta_{2015}| = 0.81$ pp for this outcome, $\bar{M} = 2$ permits an assumed post-treatment violation of up to 3.2 pp at the 2018 target and 4.9 pp at the 2019 target, against estimated effects of +32.29 pp and +34.99 pp. For **saver** and **saver_but_ips**, zero is already inside the *conventional* ($\bar{M} = 0$) confidence intervals at both dynamic targets, before any sensitivity is imposed. No positive breakdown value is established for these two outcomes, and the exercise confirms that the dynamic estimates were already inconclusive, not that they are fragile to trend violations. The complete grid is in Appendix B, Table B3.

This is a conditional-inference exercise, not a test of whether parallel trends holds: it reports what can be concluded under progressively weaker assumptions, and no result here validates or invalidates the identifying assumption. The calibration rests on a single observed pre-period movement, which the method cannot show to be representative of the confounding process and which is itself partly post-announcement. For **ips**, the permitted violations are roughly an order of magnitude smaller than the estimated effect because that benchmark is small, so surviving $\bar{M} = 2$ is a weaker endorsement than the multiplier suggests. The dynamic 2018 and 2019 targets are not the pooled primary estimand: they come from a different, 2017-inclusive sample under a less restrictive functional form, and are companions to the pooled coefficients rather than an algebraic decomposition of them.

Overall, pension-linked participation is stable across every classification variant (0.292–0.341), survives narrow-band restriction with only modest attenuation, holds under a bounded link function, and is the only outcome for which a positive trend-violation breakdown value is established. Any-saving participation is stable in magnitude (0.122–0.126, the tightest range of any outcome) but

much less secure causally. Nothing in this section addresses the single, partly post-announcement pre-period transition that is the binding constraint, and the narrow-band and dynamic evidence is mixed. Non-pension participation remains inconclusive, with nothing narrowing its interval relative to the ± 5 pp margin. Robustness here means stability across the specification and classification choices tested. It does not mean that the identification concerns on record are resolved, since none of them was targeted by, or could have been resolved by, anything in this section.

8 Income, Expenditure, and the Size of Contributions

Income and expenditure data provide a check on whether household resources moved with eligibility. These are secondary outcomes: they sit outside the primary outcome hierarchy, are estimated on the same sample with the same right-hand side, and income is never used as a covariate.

Table 7: Mechanism outcomes (N = 2,859, 55 clusters)

Outcome	Weighted	95% CI (weighted)	Cluster p (weighted)	Unweighted	Cluster p (unw.)
Log annualised consumption expenditure	0.0735 log pts (+7.63%)	[−2.72%, +19.08%]	0.145	+4.52%	0.341
Log annual disposable income	0.0024 log pts (+0.24%)	[−7.35%, +8.45%]	0.952	−0.14%	0.970
Resource residual (constant 2016 TL/year)	−5,435.1 TL	[−8,925.2, −1,944.9] TL	0.0018	−3,863.1 TL	0.041

Percentage changes are the exact transformation $100 \cdot (e^\beta - 1)$. The residual is expressed in constant 2016 Turkish lira using an annual-average consumer price index chained from TÜİK's published twelve-month-average inflation figures. The log outcomes are algebraically invariant to the deflator. The level residual is winsorised at the 1st and 99th percentiles. A fourth outcome, the residual expressed as a share of income, is reported in Appendix C. It moves in the same direction as the level residual when weighted (−6.31 pp) but is not statistically distinguishable from zero unweighted, and its significance depends on the inference method.

Disposable income shows no detectable movement (+0.24%), and neither does consumption expenditure, whose point estimate is positive (+7.63%, 95% CI [−2.72%, +19.08%]). The income–expenditure residual falls by 5,435 TL a year weighted and by 3,863 TL unweighted. The residual is estimated more precisely than either component because income and expenditure are strongly correlated (Appendix C).

A back-of-the-envelope calculation gives the size of the contributions the default produces. For civil servants, the 3% rate is charged on the pension-contribution base salary (Emeklilik Gözetim Merkezi (EGM), 2018). Under Law No. 5434 and Law No. 5510 (art. 80), that base excludes the additional payment (*ek ödeme*), the family allowance and several other allowances. According to the official salary tables, this base is about 72% of net pay for a typical civil servant, and 60–82% across titles and family situations.⁵ Each enrolled worker therefore contributes about 70 lira a month in 2016 prices. If every eligible worker in the eligible-type households of 2018–2019 stayed enrolled, contributions would come to about 1,020 lira a year per household (range 850–1,170), 1.6% of disposable income and 2.3% of consumption, with the state adding a quarter on top. At the observed public-sector stay-in rate of about 40%, the expected amount is about 410 lira a year, 0.9% of consumption. The default thus changed participation substantially while moving little money.

The fall in the residual is five to thirteen times the expected contribution, and income, where a payroll deduction would appear, does not fall, so the deduction alone cannot explain the fall. The point estimates place the fall on the expenditure side. The contributions are also well below what the design can detect: at 80% power, the minimum detectable effect is about 4,900 TL for the residual, 15% for consumption, and 12% for income. Even at full enrolment, contributions paid entirely by consuming less would lower consumption by about 2.3%, which lies inside the confidence interval. The data therefore do not show whether contributions were financed by consuming less, from other saving, or by borrowing, which the survey does not record.

The survey’s construction also limits what the residual measures. Disposable income is net of non-consumption expenditure, which includes private pension premiums, so a contribution can lower measured income through the payroll deduction, through the premium item, or both. A fall in the residual is consistent with more pension saving as well as with fewer resources. Income is recalled over the previous twelve months, whereas expenditure covers a single interview month multiplied by twelve, and the interview month is unobserved, so seasonality and within-year inflation cannot be controlled. For these reasons the residual is not treated as a measure of financial saving, although published Turkish work uses the same construction as a saving rate (Dikmen, 2021). Appendix C reports the full module.

⁵The base share is computed from the official salary coefficients and the Ministry of Treasury and Finance’s net-pay tables for 2017–2019. Stay-in rates are the shares of public-sector OKS contracts still in force, from EGM data reported in Özyerden (2026): 43% at the end of 2018 and 37% at the end of 2019. EGM’s average monthly contribution per continuing public-sector contract of workers under 45 in 2017, 82–84 TL, projected with the pay index, is within a tenth of these figures (Emeklilik Gözetim Merkezi (EGM), 2018). Shares of income and consumption are averages of each household’s contribution relative to its own income or consumption, matching the log specification of those outcomes.

9 Conclusion and Discussion

Türkiye’s 2017 automatic-enrolment reform assigned eligibility by statute: employees below age 45 at their employer’s inclusion date were enrolled by default, and those at or above it were not. Reconstructing that legal exposure from the rollout schedule and comparing eligible-type with ineligible-type households within the public sector across Household Budget Survey waves from 2015 to 2019, this paper estimates how eligibility changed household participation in saving instruments.

Legal eligibility raised reported use of the pension-linked instrument by 33.94 percentage points relative to ineligible-type households, and by 24 to 34 points across specifications. The estimate is precise under every inference method used and stable across the classification and sensitivity checks, and it is the one outcome that survives every check applied. That a default produces a large participation response is what the international evidence leads one to expect (Madrian and Shea, 2001; Choi et al., 2004; Chetty et al., 2014). The binding uncertainty is measurement, not statistics. Option 8 is a proxy, supported by the timing of its rise, the 2022 coding pattern and an official planning document, but never designated as pension-specific by TÜİK, and published work using the same item reports it under its official label (Dikmen, 2021). No amount of statistical robustness settles that question. The estimate is also measured against a comparison group whose own reported use rose.

Overall saving-instrument participation also increased, by 12.22 percentage points, and portfolio breadth by 0.323 instruments. Both are closely tied to the pension-linked instrument: the breadth increase is the instrument itself, and the participation increase is concentrated almost entirely in households whose only reported instrument is the pension-linked one. Together, the estimates describe portfolio broadening. Households appear to have added the pension-linked instrument to a largely unchanged portfolio rather than substituting it for what they already held. This is a combined reading of point estimates rather than a formally tested decomposition. The pension-only component depends most on what option 8 represents. Any-saving participation rests on the weakest pre-period evidence: its single pre-period transition is nearly half the size of the estimated effect and falls partly after the reform’s announcement, although that movement sits in the non-pension component, not in the pension-only component where the estimated effect is.

Displacement of non-pension instruments remains unresolved. At the extensive margin the limit is precision: the point estimate is close to zero wherever it is estimated, but its interval spans both sides of the ± 5 percentage-point margin, so we do not claim equivalence or displacement. For amounts the limit is scope. Non-pension participation bounds one margin only. The margins on which the recent literature locates most of the adjustment are unobserved in every wave, so a better-powered version of this design would narrow the interval on one channel and leave the others unmeasured. Of the income and expenditure measures, only the income–expenditure residual moves significantly, and it cannot be read as financial saving. The contributions involved are too small for the survey to show how they were financed.

The main limits of the design bear directly on these conclusions. Parallel trends rests on a single pre-period transition, from 2015 to 2016, and Section 7.4 reports what survives bounded violations.

The comparison group’s own reported use rose, so every estimate is relative to a group that moved, and the one alternative comparison available, with the self-employed, fails the parallel-trends test on the pension-linked outcome (Appendix D). The non-pension estimate was underpowered from the start, since its minimum detectable effect exceeds the equivalence margin. Applying the design to private-sector households would roughly halve that uncertainty, but not enough to settle the question (Appendix E). The post-period covers only 2018 and 2019, so nothing here speaks to persistence. Finally, the evidence concerns public-sector households, excluding the 12% whose classification depends on the unobserved employer category. Public-sector households show the largest rise of any household type (Appendix D, Table D2), but because private-sector rollout was staggered, that comparison does not bound the effect for others.

Participation is not the same as net saving. The two most relevant recent studies make the point in different ways: Choukhmane (2025) finds that participation effects persist while the advantage in accumulated contributions shrinks. Jensen et al. (2025), with a balance-sheet measure of saving, find that a large mandatory-saving increase largely passed through to total saving. Neither transfers directly to our setting, since both rest on quantities this survey does not contain and our post-period ends in 2019. They indicate what our participation estimates can and cannot mean, not what magnitude to expect. The reform changed how eligible public-sector households report allocating saving across instruments. Whether it changed how much they save is a question this evidence does not answer, not because the estimates are imprecise but because the margins on which net saving would move are, in these data, unobserved.

Appendix A. Treatment-Classification Algorithm

A.1 Available timing variables

Classification uses four person-level variables, none of which individually resolves eligibility. They are attained age **YAS** (whole years, all five waves); current-job tenure **CALISMA_SURE_YIL** (whole years, recorded in all five waves); employment status **FAAL_DURUM** (used with **CALISMA_SEKLI** to identify full-time, currently working members); and the social-security sector code **SAGLIK_SIGORTA_1** (13 = public/4C). The 2015 tenure column carries a different label and is not used, because every 2015 observation precedes both inclusion dates and needs no tenure to classify. No interview-month, birth-date, or employer-category variable exists in any wave (Section 3.1).

A.2 Bound derivations

Given only survey year Y , whole-year attained age A , and, where needed, whole-year tenure T , three bounds are constructed, each the widest interval consistent with the documented sources of rounding and timing uncertainty and never narrowed by an unconfirmed assumption:

(a) Age at a fixed reference date in year R_y : age at the reference date $\in [(R_y - Y + A) - 1, (R_y - Y + A) + 1]$. The interview date is unknown within Y , so birth year is bounded to $\{Y - A - 1, Y - A\}$. Projecting to a fixed date R_y and taking the widest range over all interview dates within Y yields

the ± 1 -year band. Worked example: $Y = 2018$, $A = 44$, $R_y = 2017$ (1 April) $\rightarrow$ naive estimate 43, bound $[42, 44]$.

(b) Calendar year of current-job start (computable only for $Y \geq 2016$): start year of the current job $\in [Y - T - 1, Y - T + 1]$. Two independent ± 1 -year sources of slack (the unknown interview date, and the whole-year rounding of T at the six-month boundary) are combined conservatively and are not assumed to partially cancel.

(c) Age at current-job start (used only when bound (b) indicates plausible post-rollout entry): age at hire $\in [(A - T) - 1, (A - T) + 1]$. Computed directly, to avoid compounding two separate calendar-projection uncertainty layers.

A.3 Decision procedure, per (worker, rollout scenario)

1. Missing age $\rightarrow$ *not classifiable* (missing data).
2. $Y < R_y$ (survey year strictly precedes the assumed rollout year) $\rightarrow$ *definite incumbent* with respect to that scenario, no tenure needed; eligibility from bound (a) alone. This applies to every 2015–2016 observation under both scenarios and to every 2017 observation under the January 2018 scenario. In each case the survey precedes the assumed rollout date, so the label is an eligibility-type projection from contemporaneous characteristics, not a verified legal status (Section 4.2).
3. $Y \geq R_y$ and tenure missing $\rightarrow$ *not classifiable* (missing data) (a data-availability gap, kept distinct from a genuine two-sided legal ambiguity).
4. $Y \geq R_y$ and tenure observed $\rightarrow$ compute bound (b): entirely before $R_y \rightarrow$ *definite incumbent* (bound (a)); entirely after $R_y \rightarrow$ *definite later entrant* (bound (c)); straddling $R_y \rightarrow$ *ambiguous incumbency*: compute both bound (a) and bound (c); if they agree on eligibility, the worker's eligibility is still decidable even though the governing legal clock is not; otherwise eligibility itself is *ambiguous*.

A bound with its upper end under 45 $\rightarrow$ *definitely eligible*; a lower end at or above 45 $\rightarrow$ *definitely ineligible*; otherwise $\rightarrow$ *ambiguous*.

A.4 Household aggregation

Individual statuses aggregate to the household level under a worst-case-first priority order, so that a household is labelled ineligible only if every member is definitely ineligible: (1) *eligible* (at least one definitely eligible member); (2) *ambiguous*: no eligible member, but at least one ambiguous member; (3) *not classifiable* — no eligible or ambiguous member, but at least one member missing data, or the household's one eligible-employment member turns out not to be sector-confirmed public (an edge case affecting 58 of 4,057 household-year rows, 1.4%); (4) *ineligible* (all members definitely ineligible). No ambiguous case is resolved by assumption.

Table A1: Household classification by year and rollout scenario

Scenario	Year	Definitely eligible	Definitely ineligible	Ambiguous	Not classifiable
Schedule I-IV / April 2017	2015	56.2% (434)	35.5% (274)	7.0% (54)	1.3% (10)
	2016	62.5% (594)	28.4% (270)	7.3% (69)	1.8% (17)
	2017	59.7% (483)	30.8% (249)	8.5% (69)	1.0% (8)
	2018	63.0% (480)	29.4% (224)	5.6% (43)	2.0% (15)
	2019	67.5% (516)	25.3% (193)	6.2% (47)	1.0% (8)
Other public / January 2018	2015	52.2% (403)	39.0% (301)	7.5% (58)	1.3% (10)
	2016	59.1% (561)	31.8% (302)	7.4% (70)	1.8% (17)
	2017	56.9% (460)	35.4% (286)	6.8% (55)	1.0% (8)
	2018	59.4% (453)	31.8% (242)	6.8% (52)	2.0% (15)
	2019	63.4% (484)	28.9% (221)	6.7% (51)	1.0% (8)

Table A2: Pooled classification, 2015–2019 (N = 4,057)

Scenario	Definitely eligible	Definitely ineligible	Ambiguous	Not classifiable
Schedule I-IV / April 2017	61.8% (2,507)	29.8% (1,210)	7.0% (282)	1.4% (58)
Other public / January 2018	58.2% (2,361)	33.3% (1,352)	7.0% (286)	1.4% (58)

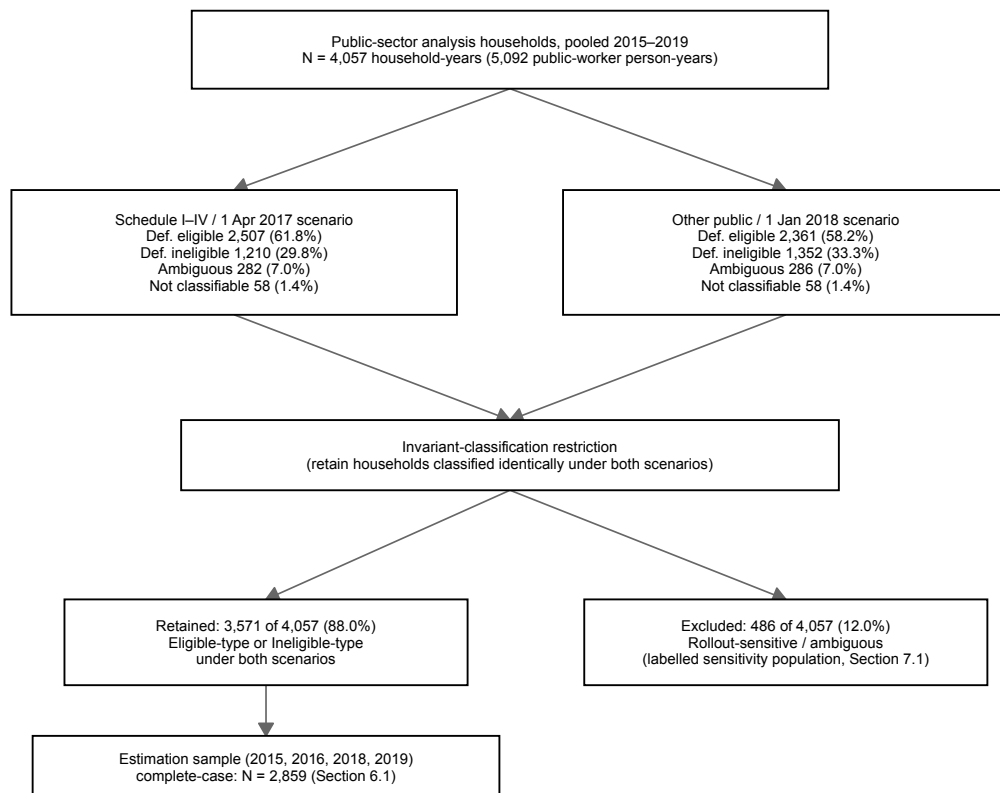
A.5 By-year $\times$ by-scenario classification tables

Households classified identically under both scenarios (Tables A1 and A2): 92.9% pooled (2015: 92.5%; 2016: 93.2%; 2017: 92.6%; 2018: 94.1%; 2019: 92.1%). The remaining 7.1% change classification depending on which rollout date is assumed, concentrated in years and scenarios where the survey year is close to the assumed rollout year.

Figure A1 traces the flow from the full public-sector analysis population through the two scenario classifications to the invariant-classification restriction (Section 4.2).

A.6 Comparison against the single-date rule of Section 4.2

A residual 0.1% "other" category, pooled, reflects households for which the single-date indicator is itself missing. The single-date rule agrees with the strict definite-status classification for the large majority of households (89–91%) and disagrees outright for a small minority (0.7–2.1%) (Table A3). The remainder (about 8.4–8.5%) are cases the interval method, which is deliberately conservative, cannot resolve to a definite status under either date.



Source: Authors' calculations from TÜİK Household Budget Survey, 2015–2019. All counts are household-years, pooled across waves; see Section 4 for the classification rule and Appendix A for full detail.

Figure A1: Treatment-classification flow

Table A3: Single-date age rule versus the interval-based invariant classification

Scenario	Agree (exposed)	Agree (unexposed)	Total agree	Disagree	Ambiguous/unclassifiable under the interval rule
Schedule I–IV / April 2017	61.6% (2,499)	29.2% (1,184)	90.8%	0.7% (29)	8.4% (340)
Other public / January 2018	58.1% (2,358)	31.2% (1,266)	89.3%	2.1% (84)	8.5% (344)

A.7 Year fixed effects and the Post indicator

Equation (1) includes a full set of survey-year fixed effects and no separate Post_t term. The estimated model also includes Post_t . Because the estimation sample contains only 2015, 2016, 2018 and 2019, Post_t equals the sum of the 2018 and 2019 year indicators, so one of the two is dropped and its effect is absorbed by Post_t . This affects only the year and Post coefficients. The $\text{Eligible}_h \times \text{Post}_t$ column is not a linear combination of the other regressors in either form, so the two parameterisations span the same column space and give identical fitted values, residuals, estimates of δ , and standard errors.

Appendix B. Full Robustness and Identification-Sensitivity Results

B.1 Classification-robustness detail

Wild-bootstrap p-values track the cluster-robust p-values closely for every variant and outcome (Table B1). None changes the significance conclusion at the 5% level. The symmetric-ambiguity sample additionally excludes any household containing an ambiguously classified worker in either group (97 further household-years dropped from the 3,571 invariant-classification sample, pooled over five years). The single-date rule holds timing, covariates, weights, and clustering fixed and varies only the classification rule ($N = 3,241$, 58 cohort clusters, since every household receives a definite status under this rule by construction).

Table B1: Per-variant coefficients, standard errors, and cluster counts (weighted)

Variant	N	Clusters	ips	saver_but_ips	saver	savings_count
Primary	2,859	55	0.3394 (SE 0.0330)	−0.0050 (SE 0.0486)	0.1222 (SE 0.0503)	0.3233 (SE 0.0813)
Symmetric-ambiguity sample	2,782	55	0.3406 (SE 0.0333)	−0.0015 (SE 0.0486)	0.1254 (SE 0.0506)	0.3292 (SE 0.0816)
Single-date classification rule	3,241	58	0.2917 (SE 0.0337)	0.0227 (SE 0.0353)	0.1260 (SE 0.0355)	0.3033 (SE 0.0618)

B.2 Logit average cross-difference companion (unweighted, $N = 2,859$)

All three preserve the qualitative significance conclusions of the unweighted linear estimates (Table B2). The non-pension estimate changes sign but remains close to zero. `savings_count` is excluded because it is a count, not a binary outcome. This is the sample average of the object A_i and Norton define, not the observation-level effect, which can vary in sign across observations.

Table B2: Sample-averaged interaction effect in the sense of Ai and Norton (2003)

Outcome	AME at Post = 0	AME at Post = 1	Cross-difference	SE	p	95% CI	Unweighted LPM
ips	0.0032	0.3095	0.3063	0.0273	3.28×10^{-29}	[0.253, 0.360]	0.3394
saver_but_ips	0.0534	0.0501	-0.0033	0.0391	0.932	[-0.080, 0.073]	0.0022
saver	0.0541	0.1790	0.1248	0.0381	0.00106	[0.050, 0.200]	0.1330

AME is the average marginal effect of Eligible on the outcome, evaluated at Post = 0 and at Post = 1. SE is the delta-method standard error of the cross-difference between them, as reported by `marginaleffects::avg_slopes()`.

Fitting the same logit specification with `FAKTOR` as binomial prior weights did not converge, producing coefficient magnitudes of order 10^{14} – 10^{15} for all three outcomes. Removing the weights while holding the formula and data fixed converges cleanly. No alternative estimator (Firth logistic regression, survey-design GLM) was substituted. The logit companion is reported unweighted only, compared against the unweighted LPM.

B.3 Complete $\bar{M}$ grid of robust confidence sets

For `ips`, both targets exclude zero at every evaluated $\bar{M}$. The sample breakdown value is strictly greater than 2 for both. For `saver` and `saver_but_ips`, zero is already inside the conventional ($\bar{M} = 0$) interval at both targets, before any sensitivity is imposed. The grid confirms inconclusiveness and reveals no new fragility.

Table B3: Rambachan–Roth relative-magnitude robust confidence sets, cohort-clustered covariance

Outcome	Target	Conventional CI ($\bar{M} = 0$)	$\bar{M} = 0.5$	$\bar{M} = 1$	$\bar{M} = 1.5$	$\bar{M} = 2$	Zero excluded at every $\bar{M}$?
ips	2018	[22.77, 41.82] pp	[21.41, 41.97]	[18.94, 43.68]	[16.08, 45.96]	[12.85, 48.82]	Yes
ips	2019	[25.25, 44.72] pp	[22.09, 46.02]	[17.22, 48.36]	[11.77, 51.08]	[5.94, 56.34]	Yes
saver	2018	[−2.06, 22.26] pp	[−11.54, 28.56]	[−25.40, 40.95]	[−40.22, 55.05]	[−54.80, 69.63]	No — never
saver	2019	[−5.61, 19.76] pp	[−21.16, 32.06]	[−42.45, 52.59]	[−64.76, 74.39]	[−87.06, 96.69]	No — never
saver_but_ips	2018	[−13.27, 9.25] pp	[−23.51, 14.73]	[−37.68, 27.11]	[−52.30, 41.28]	[−67.15, 55.68]	No — never
saver_but_ips	2019	[−18.94, 7.01] pp	[−34.35, 18.79]	[−55.34, 39.27]	[−77.37, 60.78]	[−99.41, 82.82]	No — never

B.4 Dynamic-companion coefficient vectors

Table B4: Event-study coefficients, 2016 reference, cohort-clustered SE, in percentage points (N = 3,567, 56 clusters)

Outcome	2015	2017	2018	2019
ips	−0.81 (1.89)	−0.06 (2.29)	+32.29 (4.75)	+34.99 (4.86)
saver	−7.62 (4.54)	−8.25 (6.52)	+10.10 (6.07)	+7.08 (6.33)
saver_but_ips	−7.26 (4.68)	−8.02 (6.67)	−2.01 (5.62)	−5.97 (6.47)

Standard errors in parentheses. Figure 3 (Section 7.4) renders this table as confidence intervals.

Appendix C. Income and Expenditure Mechanism Module, in Full

C.1 All four pre-specified outcomes, weighted and unweighted

O4's significance depends on both weighting and inference method, and the paper does not describe it as robust (Table C1). Weighted, the cluster p-value is 0.026 and the wild-bootstrap p-value 0.030, but the HC1 p-value is 0.070. For this outcome the cohort-clustered standard error (0.0283) is smaller than the HC1 one (0.0348), the reverse of every primary outcome, where clustering inflates the standard error by 10–20%. Unweighted, O4 is not significant under any method (cluster p = 0.121).

Table C1: Mechanism outcomes, complete (N = 2,859, 55 clusters)

Outcome	Weighted coefficient	Cluster SE	Cluster p	Wild-boot. p	Unweighted coefficient	Cluster SE (unw.)	Cluster p (unw.)
O1: log annualised expenditure	0.0735 (+7.63%)	0.0504	0.145	0.161	0.0442 (+4.52%)	0.0465	0.341
O2: log disposable income	0.0024 (+0.24%)	0.0393	0.952	0.952	-0.0014 (-0.14%)	0.0382	0.970
O3: real resource residual, winsorised (constant 2016 TL/yr)	-5,435.08	1,740.83	0.0018	0.0016	-3,863.09	1,891.45	0.041
O4: residual rate, winsorised (share of income)	-6.31 pp	2.83 pp	0.026	0.030	-4.80 pp	3.09 pp	0.121

Percentage changes for O1/O2 use the exact transformation $100 \cdot (e^{\beta} - 1)$. O3/O4 are winsorised at the 1st/99th percentiles of their own distribution, computed once on the pooled N = 2,859 sample before estimation (O3 thresholds [-78,658.15, 100,710.95] TL; O4 thresholds [-1.3696, 0.7073]; 29 households affected at each tail, for each outcome).

Table C2: Official TÜİK annual-average CPI series used to deflate O3 (base 2016 = 100)

Year	Official index
2015	92.78
2016	100.00
2017	111.14
2018	129.29
2019	148.92

C.2 Winsorised and unwinsorised residual estimates

Table 7 reports $-5,435.1$ TL, the pre-specified winsorised weighted O3 coefficient. A second number, $-4,445$ TL, appears only in this appendix. It is the unwinsorised weighted O3 coefficient, computed after estimation as a diagnostic, and it is used only to explain why the residual is estimated more precisely than either of its components. By OLS linearity, the unwinsorised residual’s coefficient equals the exact difference of its two unwinsorised components:

Component	Unwinsorised real-TL coefficient	Cluster SE
Income (O2, real levels)	+228.0 TL	2,915.2
Annualised expenditure (O1, real levels)	+4,673.4 TL	3,428.4
Difference (= unwinsorised O3)	$-4,445.4$ TL	1,946.6

The difference’s standard error (1,946.6) is smaller than either component’s (2,915.2 and 3,428.4) because household income and expenditure move together strongly (the implied correlation between the two component estimators is $\rho \approx 0.82$). Differencing therefore cancels the common household-scale variation that makes each component noisy. This follows from $\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2\text{Cov}(A, B)$. It is not an economic mechanism.

The $-4,445$, $+228$ and $+4,673$ TL figures are not pre-specified outcomes. They were computed after the primary O3 result was known and are reported only to explain the precision pattern. Winsorisation, fixed in the specification before any monetary result existed, moves the coefficient from $-4,445.4$ TL ($p = 0.023$) to the reported $-5,435.1$ TL ($p = 0.0018$), a 22.3% increase in magnitude. Both are significant at 5%.

C.3 Deflator construction

The index is chained from TÜİK’s December press-bulletin figures for the twelve-month-average year-on-year change: 2016 +7.78%, 2017 +11.14%, 2018 +16.33%, 2019 +15.18% (Table C2). O1, O2 and O4 are algebraically invariant to the deflator: a common annual index is absorbed by the year fixed effects for the log outcomes and cancels in O4’s same-year ratio.

C.4 Sample and negative-residual share

All four outcomes are estimated on the full $N = 2,859$ mechanism sample (55 cohort clusters). YILKULGEL and HARCAMA have zero missing and zero non-positive values in the merged panel, so no household is excluded for any outcome. Before winsorisation, 804 of 2,859 households (28.1%) show income below annualised expenditure ($O3 < 0$), descriptive context, not a treatment effect.

Appendix D. Alternative Specifications and the Cross-Sector Comparison

This appendix reports two alternative analyses, distinct from the robustness checks of Appendix B because each rests on a different sample or comparison group rather than a variant of the primary design. The first (D.1) is an alternative implementation of the within-public-sector comparison. It uses a single age cutoff evaluated at one representative date instead of the interval-based, dual-scenario classification of Section 4. It also uses a different pre-period (2015–2017 pooled, not 2015–2016 with 2017 excluded) and a different covariate set (household income included as a control, not excluded as a plausible mediator). The second (D.2) is the public-versus-self-employed cross-sector comparison, used only for the parallel-trends diagnostic cited in Section 5.3.

D.1 Headline comparison against an alternative specification

The alternative specification’s estimates are as reported in the thesis (Table D1). Any-saving participation is the closest match between the two specifications despite the joint differences in sample construction, timing, and covariates. Pension-linked participation is larger in the present design (+5.2 pp). The gap reflects the bundle of design differences, most plausibly the exclusion of income as a covariate and the exclusion of the most rollout-ambiguous households from the invariant sample, and no decomposition of it is attempted. Non-pension participation is near zero and non-significant under both specifications. Portfolio breadth is somewhat larger in the present design.

Table D1: Present estimates versus the alternative specification

Outcome	Present design (weighted, cohort-clustered SE)	Alternative specification (HC1, $N = 4,049$)
Pension-linked	0.339 (SE 0.033)	0.2869 (SE 0.0287)
Non-pension	−0.005 (SE 0.049)	0.0247 (SE 0.0368)
Any-saving	0.122 (SE 0.050)	0.1285 (SE 0.0352)
Portfolio breadth	0.323 (SE 0.081)	0.289 (SE 0.0562)

D.2 The cross-sector parallel-trends test cited in Section 5.3

The public-versus-self-employed comparison for workers aged 45 and over, cited in Section 5.3 as a reason against adopting that comparison, fails the parallel-trends test on exactly the pension-linked outcome ($F = 3.15$, $p = 0.043$), the only failure among nine comparison–outcome cells tested. It uses its own classification rule, pre-period, and covariate set, distinct from both the primary design and the alternative specification of D.1. It is reported here as evidence on the credibility of the cross-sector comparison itself, not as a result of the present design, and it is not recomputed under the present specification.

D.3 Participation by household type

Table D2 reports weighted shares of pension-linked, any-saving and non-pension participation by household type and period for all 59,102 household-years, using the member-level sector codes of Section 3.1 with no age or eligibility restriction. Household types are: public (all coded members public, at least one full-time salaried worker); private (all coded members private, at least one full-time salaried worker); and mixed (members from more than one of public, private and self-employed). The other three are self-employed (coded members self-employed only); no identified worker (coded members but no full-time salaried worker, largely retired households); and other-only (no member coded 4A, 4B or 4C). The 2019 code list for the health-insurance variable differs for categories other than 4A–4C, so the other-only group’s composition differs in that year. Pension-linked participation rises by 40.8 percentage points among public households, 20.6 among mixed, 14.3 among private, and under 6 elsewhere. Private and mixed households were still being phased in through 2019, so the ranking reflects exposure at least as much as response, and it is not a bound on the effect for any group.

Table D2: Saving-instrument participation by household type, weighted shares, pre and post

Household type	N (2015–2019)	Pre	Post	Change
<i>Panel A. Pension-linked participation</i>				
Public	4,057	2.3%	43.1%	+40.8 pp
Mixed	6,665	1.5%	22.1%	+20.6 pp
Private	19,772	0.8%	15.2%	+14.3 pp
Self-employed	8,818	0.3%	6.0%	+5.8 pp
No identified worker	12,212	0.4%	4.2%	+3.9 pp
Other-only	7,578	0.1%	2.3%	+2.2 pp
<i>Panel B. Any-saving participation</i>				
Public	4,057	39.8%	69.2%	+29.4 pp
Mixed	6,665	36.2%	56.6%	+20.4 pp
Private	19,772	19.0%	38.5%	+19.5 pp
Self-employed	8,818	26.7%	41.4%	+14.7 pp
No identified worker	12,212	23.0%	34.9%	+11.9 pp
Other-only	7,578	9.1%	16.9%	+7.8 pp
<i>Panel C. Non-pension participation</i>				
Public	4,057	38.8%	51.3%	+12.5 pp
Mixed	6,665	35.5%	46.9%	+11.3 pp
Private	19,772	18.5%	29.4%	+10.8 pp
Self-employed	8,818	26.7%	38.9%	+12.2 pp
No identified worker	12,212	22.8%	32.4%	+9.6 pp
Other-only	7,578	9.0%	15.8%	+6.7 pp

Weighted by FAKTOR. Pre = 2015–2016 pooled, post = 2018–2019 pooled; 2017 excluded. All 59,102 household-years, no age or eligibility restriction. N counts household-years in all five waves; the shares use the pre- and post-period waves stated.

Appendix E. A private-sector extension: design and ex ante power

A natural way to add power is to apply the design to private-sector households. Private employers were brought into OKS by firm size between January 2017 and January 2019, and the survey records neither firm size nor the interview month, so the timing cannot be matched household by household. By 1 January 2019, however, all employers with five or more employees were included, so 2019 can serve as a post-reform year, with 2015–2016 as the pre-period and 2017 and 2018 as transition years. Eligibility can then be defined by age alone. A worker is eligible if under 45 even at the last inclusion date, 1 January 2019, and ineligible if 45 or older already at the first, 1 January 2017. The cohorts in between are excluded. This rule applies to incumbents and later hires alike, so tenure is not needed.

We fixed this design before computing its power, and used only the 2015–2016 outcomes and the 2019 sample composition; no post-reform outcome was used. Applied to the public-sector design, the same method reproduces the paper’s own ex ante power figures exactly (Table E1).

The private design roughly halves the uncertainty of the non-pension estimate. Its expected

Table E1: Private-sector extension: ex ante power, non-pension participation, weighted

Design	MDE at 80% power (pp)	Expected 95% interval half-width (pp)	Power of the equivalence test (± 5 pp), true effect 0	Power to detect a 5 pp displacement
Public design (paper)	12.3 (13.0)	8.6 (9.1)	0% (0%)	21% (19%)
Private design	6.7 (7.2)	4.7 (5.0)	10% (0%)	55% (49%)
Private design, counting 2018 as post	5.2 (5.7)	3.7 (4.0)	53% (37%)	76% (68%)

FAKTOR-weighted. Outcome is non-pension participation. Each cell gives the estimate without an adjustment for clustering, followed in parentheses by the estimate with an adjustment for clustering by birth cohort. Rows 1 and 2 use the 2015, 2016 and 2019 sample composition; row 3 adds 2018. Source: authors’ calculations from TÜİK Household Budget Survey, 2015–2016 outcomes and 2019 sample composition.

95% interval is still about as wide as the ± 5 -point equivalence margin, so it could not establish equivalence, and it would detect a displacement of 5 points only about half the time. The binding constraint is the small ineligible group in the single post year (551 households in 2019). Counting 2018 as a post year raises power, but it treats employers with 5 to 49 employees as enrolled before they were. Plausible displacement effects are also smaller in the private sector: its non-pension base is about 19%, against about 42% in the public sector, and fewer private-sector contracts stayed in force (30% at the end of 2018, against 43% in the public sector). For these reasons the extension is not pursued in this paper.

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